DISTRICT PANCHAYATH KASARAGOD

EQIP 2026

Educational Quality Improvement Programme for class -X

SOCIAL SCIENCE II

ENGLISH MEDIUM

Standard -X

QUESTION POOL



DISTRICT INSTITUTE OF EDUCATION AND TRAINING (DIET)
KASARAGOD

EQIP 2025-26

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ആശംസ

വികേന്ദ്രീകൃത ആസൂത്രണത്തിലൂടെയും നിർവ്വഹണത്തിലൂടെയും കേരളത്തിലെ ആരോഗ്യവിദ്യാഭ്യാസ മേഖലകളെ ദേശീയ തലത്തിൽ ഒന്നാമതെത്തിക്കാൻ നമുക്ക് കഴിഞ്ഞിട്ടുണ്ട്. ഈ നേട്ടങ്ങൾ കൈവരിക്കാൻ പ്രാദേശിക ഭരണകൂടങ്ങൾ സ്തുത്യർ ഹമായ പങ്കുവഹിച്ചു. ദേശീയ സംസ്ഥാനതല പഠനങ്ങൾ നമ്മുടെ കുട്ടികളുടെ പഠന നിലവാരം ഇനിയും ഉയരേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറെ വർഷങ്ങളായി കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ പൊതുവിദ്യാഭ്യാസ വകുപ്പും കാസറഗോഡ് ഡയറ്റും ഒത്തു ചേർന്ന് നടപ്പാക്കുന്ന **'എക്വിപ്പ്' 2025-26 (EQIP 2025-26)** പഠനപരിപോഷണ പരിപാടിക്ക് എല്ലാ പിന്തുണയും ഉറപ്പുതരുന്നു. പൊതുപരീക്ഷകളെ അഭിമുഖീകരിക്കുന്ന പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രവർത്തനങ്ങളുടെ മികവിന്റെ അടയാളമായി മാറുകയാണ് വാർഷിക പരീക്ഷകൾ. അറിവിന്റെ തെളിമയോടെ ഓരോ വിദ്യാർത്ഥിക്കും പരീക്ഷ എഴുതാൻ കഴിയണം. വിദ്യാർത്ഥികളുടെ ജീവിതത്തിലെ ഏറ്റവും പ്രധാനപ്പെട്ട പരീക്ഷകൾക്ക് വേണ്ടി തയ്യാറാക്കിയ ചോദ്യബാങ്കിന് എല്ലാവിധ ആശംസകളും നേരുന്നു. നന്നായി പഠിക്കുക. പരീക്ഷയെ സധൈര്യം നേരിടുക. തളരാതെ മുന്നോട്ട്. വിജയം നിങ്ങളോടൊപ്പമുണ്ട്. ആശംസകൾ.

ശ്രീമതി ബേബി ബാലകൃഷ്ണൻ
ജില്ലാ പഞ്ചായത്ത് പ്രസിഡന്റ്
കാസറഗോഡ്



ആശംസ

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസ മേഖലയെ ശക്തിപ്പെടുത്തുന്നതിന് നിരവധി പ്രവർത്തനങ്ങളാണ് നടന്നുവരുന്നത്. പൊതുവിദ്യാഭ്യാസ മേഖലയെ പൂർവ്വാധികം കരുത്തോടെ നാം മുന്നോട്ട് നയിക്കുകയാണ്. ഈ ഘട്ടത്തിലാണ് കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തും, പൊതുവിദ്യാഭ്യാസ വകുപ്പും, വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രവും (DIET) പത്താം ക്ലാസ് വിദ്യാർത്ഥികളുടെ പഠന വിടവുകൾ പരിഹരിക്കുന്നതിനും ആത്മവിശ്വാസത്തോടെ പൊതുപരീക്ഷയെ നേരിടാൻ അവരെ പ്രാപ്തരാക്കുന്നതിനും വേണ്ടി പഠനപരിപോഷണ സാമഗ്രി തയ്യാറാക്കിയിട്ടുള്ളത്. ഈ പദ്ധതിയുടെ ഭാഗമായി ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ചുള്ള ചോദ്യശേഖരമാണ് (Question Pool) ഈ അക്കാദമിക വർഷം തയ്യാറാക്കിയിട്ടുള്ളത്. കുട്ടികളുടെ അക്കാദമിക് മികവ് ഉറപ്പുവരുത്തിക്കൊണ്ട് മികച്ച ഗ്രേഡുകൾ നേടാൻ അവരെ സജ്ജമാക്കാൻ **എക്വിപ് 2025-26 (EQIP 2025-26)** എന്ന പേരിൽ തയ്യാറാക്കിയ ഈ പദ്ധതിക്ക് കഴിയട്ടെയെന്ന് ആശംസിക്കുന്നു.

സ്നേഹപൂർവ്വം

അഡ്വ. സരിത എസ്.എൻ.

ആരോഗ്യ-വിദ്യാഭ്യാസ സ്ഥിരം സമിതി അധ്യക്ഷ,

ജില്ലാ പഞ്ചായത്ത്, കാസറഗോഡ്



ആശംസ

ജില്ലയിലെ അക്കാദമിക പ്രവർത്തനങ്ങളെ ഏകോപിപ്പിച്ച് മുന്നോട്ട് നയിക്കുന്നതിന്റെ പ്രധാന ഉത്തരവാദിത്തം കാസറഗോഡ് ജില്ലാ വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രത്തിന്റെ (DIET) നേതൃത്വത്തിൽ കാലങ്ങളായി ചെയ്തുവരുന്നുണ്ട്. പ്രീ-പ്രൈമറി തലം മുതൽ ഹയർസെക്കണ്ടറി തലംവരെയുള്ള മേഖലകളിൽ വിവിധങ്ങളായ പദ്ധതികൾ ആസൂത്രണം ചെയ്യാനും നിർവഹിക്കാനും ഡയറ്റിന് ഇതിനകം കഴിഞ്ഞിട്ടുണ്ട്. പത്താം ക്ലാസ്സിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറച്ചു വർഷങ്ങളായി ഡയറ്റിന്റെ നേതൃത്വത്തിൽ വിദ്യാഭ്യാസ വകുപ്പ് നടപ്പിലാക്കുന്ന പദ്ധതിയാണ് **എക്വിപ് (Educational Quality Improvement Programme).** അതാതു വർഷത്തെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഗണിച്ചുകൊണ്ടാണ് പ്രവർത്തനങ്ങൾ ആസൂത്രണം ചെയ്യുന്നത്. പത്താംതരം പാഠപുസ്തകവും വിലയിരുത്തൽ രീതിയും മാറിയ ഘട്ടത്തിൽ ആണല്ലോ നമ്മൾ. അതുകൊണ്ട് ഈ പദ്ധതിയുടെ ഭാഗമായി പത്താംതരത്തിലെ ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ പുതിയ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ച് ചോദ്യശേഖരം (Question Pool) പരിചയപ്പെടുത്താനാണ് ഇപ്രാവശ്യം ഉദ്ദേശിക്കുന്നത്. ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ഡയറ്റിന്റെ അക്കാദമിക സഹായത്തോടെ മലയാളത്തിലും ഇംഗ്ലീഷിലും കന്നഡയിലും പത്താം തരത്തിലെ ഐ.ടി. ഒഴികെയുള്ള വിഷയങ്ങളിൽ ചോദ്യശേഖരം തയ്യാറാക്കി നൽകാനാണ് ഉദ്ദേശിക്കുന്നത്. ഈ അധ്യയനവർഷം പത്താംതരം പരീക്ഷ എഴുതുന്ന മുഴുവൻ കുട്ടികൾക്കും ഈ പദ്ധതിയുടെ പ്രയോജനം ലഭിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു. വ്യത്യസ്ത പഠനശേഷിയും പഠനമികവുമുള്ള എല്ലാ വിഭാഗം കുട്ടികൾക്കും ഈ സാമഗ്രി പ്രയോജനപ്പെടട്ടെ എന്ന് ആശംസിക്കുന്നു. ഒരു കൂട്ടം അധ്യാപകരുടെ ആത്മാർത്ഥമായ പ്രവർത്തനമാണ് ഈ സാമഗ്രി വികസിപ്പിക്കാൻ സഹായകമായത്. അധ്യാപകരുടെ ആത്മാർത്ഥമായ പിന്തുണയും പ്രോത്സാഹനവും അനിവാര്യമായ ഈ ഉദ്യമത്തിൽ എല്ലാവരുടെയും സഹായസഹകരണങ്ങൾ തുടർന്നും പ്രതീക്ഷിച്ചുകൊണ്ട് ഏവർക്കും വിജയാശംസകൾ നേരുന്നു. ആശംസകളോടെ,

ഡോ. രഘുരാമ ഭട്ട് കെ.
പ്രിൻസിപ്പാൾ
ഡയറ്റ് കാസറഗോഡ്



ആമുഖം

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസം ശക്തിപ്പെടുത്തുന്നതിന് വ്യത്യസ്തങ്ങളായ നിരവധി പ്രവർത്തനങ്ങൾ നടന്നുവരികയാണ്. അതേസമയം ദേശീയ-സംസ്ഥാന പഠനങ്ങൾ നമ്മുടെ ജില്ലയിലെ കുട്ടികളുടെ പ്രകടനം ഇനിയും മെച്ചപ്പെടേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. ഈ പശ്ചാത്തലത്തിലാണ് ജില്ലയിലെ പൊതുവിദ്യാലയങ്ങളിൽ നിന്ന് **2025-26** അധ്യയന വർഷം എസ്.എസ്.എൽ.സി. പരീക്ഷകൾ അഭിമുഖീകരിക്കുന്ന കുട്ടികൾക്ക് പഠനപിന്തുണ നൽകുന്നതിന് ജില്ലാ പഞ്ചായത്തിന്റെയും പൊതുവിദ്യാഭ്യാസ വകുപ്പിന്റെയും സംയുക്താഭിമുഖ്യത്തിൽ വ്യത്യസ്ത വിഷയങ്ങളിൽ പഠനസാമഗ്രികൾ തയ്യാറാക്കേണ്ടതിന്റെ ആവശ്യകത ജില്ലാതല ഉന്നതാധികാര യോഗങ്ങളിൽ ചർച്ചചെയ്യപ്പെട്ടത്. ഇതിന്റെ അടിസ്ഥാനത്തിൽ ഡയറ്റ് കാസറഗോഡിന്റെ അക്കാദമിക നേതൃത്വത്തിൽ ജില്ലയിലെ മികച്ച അധ്യാപകരെ ഉൾപ്പെടുത്തിക്കൊണ്ട് പത്താംതരത്തിൽ ഐ.ടി. ഒഴിച്ചുള്ള എല്ലാ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിലനുസരിച്ച് ചോദ്യശേഖരം **(Question Pool)** തയ്യാറാക്കിയിരിക്കുകയാണ്. ജില്ലയിൽ നിന്നും പൊതുപരീക്ഷയെ അഭിമുഖീകരിക്കുന്ന മുഴുവൻ എസ്.എസ്.എൽ.സി, വിദ്യാർത്ഥികൾക്കും ആത്മവിശ്വാസം വളർത്തുന്നതിനും ഉന്നത വിജയം നേടുന്നതിനും ഈ ഉദ്യമം സഹായകമാകട്ടെയെന്ന് ആത്മാർത്ഥമായി ആഗ്രഹിക്കുന്നു. ഈ പദ്ധതിയെ നെഞ്ചേറ്റിയ പ്രിയപ്പെട്ട അധ്യാപക സുഹൃത്തുക്കൾക്ക് ഈ പുസ്തകത്തെ ഫലപ്രദമായി ഉപയോഗിക്കാൻ കഴിയട്ടെ.

എല്ലാവർക്കും വിജയാശംസകൾ.

ശ്രീ. മധുസൂദനൻ ടി.വി.
ജില്ലാ വിദ്യാഭ്യാസ ഉപഡയറക്ടർ
കാസറഗോഡ്

SOCIAL SCIENCE II

English Medium

Chapter 1

WEATHER AND CLIMATE

1. Find and write the incorrect statement from the following:

   a) Factors such as atmospheric temperature, pressure, wind, and humidity experi enced in an area are called weather elements.

   b) The conditions like temperature, atmospheric pressure, winds, humidity, and precipitation experienced in an area over a short period are called weather.

   c) The weather condition experienced in an area over a short period is called climate.

   d) The climate of an area is determined by considering the weather conditions experienced over a vast geographical area for approximately 35 to 40 years.

2. Prepare a note for a seminar presentation on the topic: 'The Influence of Climatic Factors on Human Life'.

3. Explain the difference between weather and climate.

4. Find and write the correct statement(s) related to solar radiation from the following:

   i) The earth's surface does not get heated through solar radiation.

   ii) A part of the solar radiation coming from the sun to the earth is reflected by atmospheric substances.

   iii) Solar radiation passing through the atmosphere heats the atmosphere because it is long-wave radiation.

   iv) The earth's surface gets heated through solar radiation.

   a) i, iii correct b) i, iv correct
   c) ii, iv correct d) ii, iii correct

5. Explain the following concepts:

   a) Solar Radiation

   b) Insolation

   c) Terrestrial Radiation

   d) Heat Budget

6. The earth's surface gets heated through solar radiation. Subsequently, heat transfer occurs from the earth's surface to the atmosphere through various heating processes. Explain the main heating processes.

   Or

   "The atmosphere is primarily heated from the earth's surface." Justify this statement through the atmospheric heat transfer processes.

7. Find the incorrect pair:

   i) Advection - Heat spreads to the lower levels of the atmosphere near the heated earth's surface.

   ii) Conduction - Heat spreads horizontally through winds.

   iii) Convection - Heat spreads to greater heights as the heated air rises up from the earth.

   iv) Radiation - As the earth's surface gets heated, energy is emitted from the earth's surface as long waves.

   a) i, ii, iii b) i, ii c) iii, iv d) i, iv

8. Do we receive the same amount of solar energy throughout the day? Why do weather observers consider the temperature recorded at 2 p.m. as the maximum atmospheric temperature for a day?

9. Select the correct pairs:

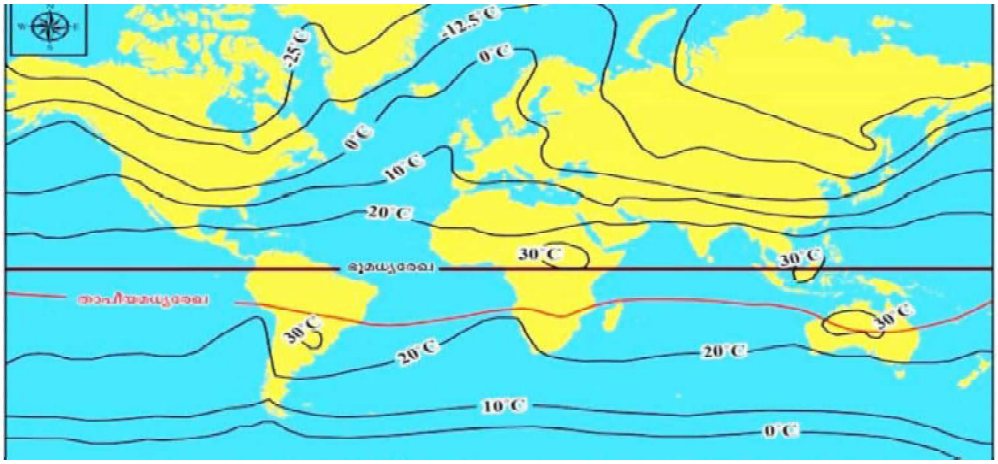
a)	Atmospheric Temperature	Thermometer, Maximum-Minimum Thermometer
b)	Atmospheric Pressure	Anemometer, Aneroid Barometer
c)	Atmospheric Humidity	Hygrometer, Wet and Dry Bulb
d)	Wind Speed & Direction	Barometer, Wind Vane

   i) a, b correct ii) a, d correct iii) a, c correct iv) c, d correct

10. How is the Diurnal Range of Temperature and the Daily Mean Temperature calculated using the maximum and minimum temperatures of an area for a day?

11. Calculate the Diurnal Range of Temperature and the Daily Mean Temperature, assuming the maximum temperature of an area is 34^0C and the minimum temperature is 26^0C.

12. Explain the difference between Isotherms and Isobars.

13. 

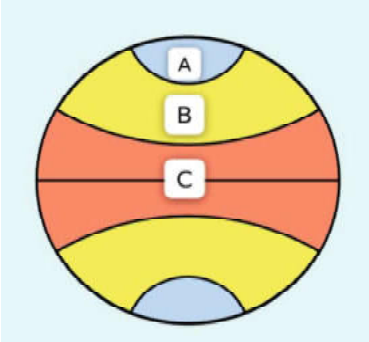
Observe the image provided above. (Global Temperature Distribution in degrees Celsius for January).

a) Is the arrangement of isotherms uniform everywhere?

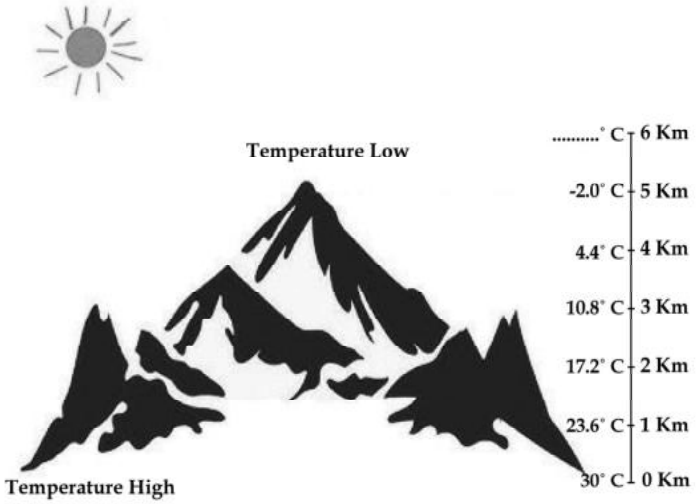
b) Why do the isotherms noticeably bend where continents and oceans meet?

c) What could be the reason why the isotherms in the Southern Hemisphere are almost parallel to the latitudes, compared to the Northern Hemisphere?

14. The image provided below is related to Thermal Zones Observe the image and identify the thermal zones marked A, B, and C, and specify the latitudes between which they are situated.



15. Observe the image and find the answers to the questions given below:



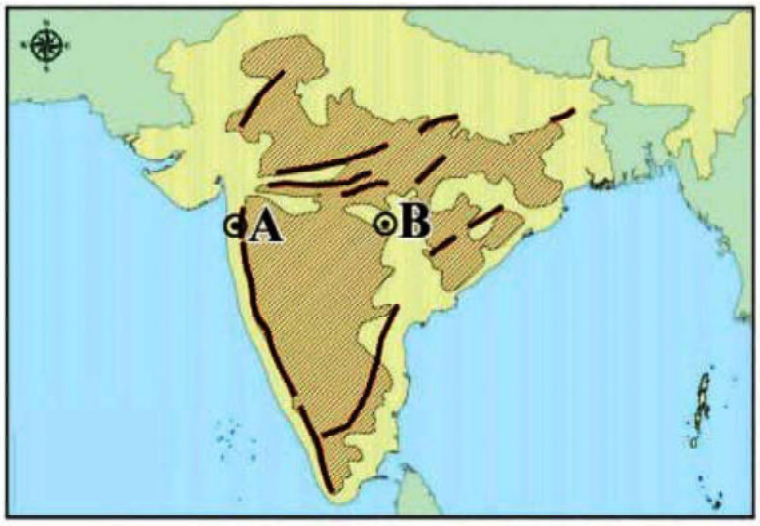
a) What change occurs in the temperature as the height increases?

b) What is the change in temperature for every kilometer increase in height?

c) What is the name of this change (phenomenon) in atmospheric temperature?

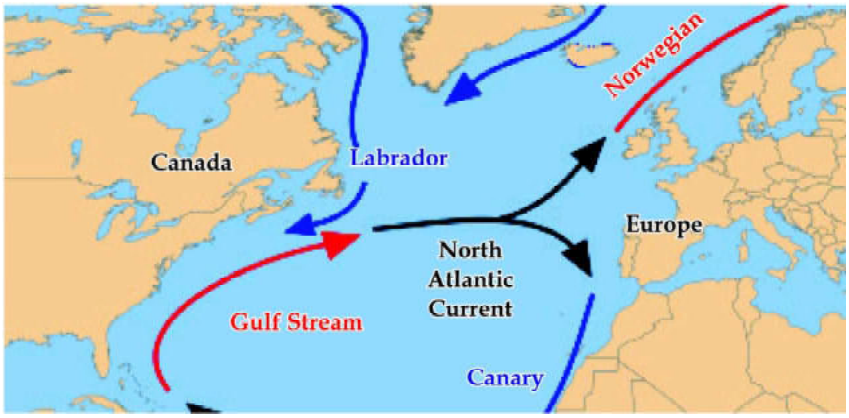
16. Why do high-altitude regions like Ooty, Munnar, and Kodaikanal generally experience low temperatures?

17. The given image indicates the Diurnal Temperature In the map, where would the Diurnal Range of Temperature be greater, at the places marked A or B? Why?

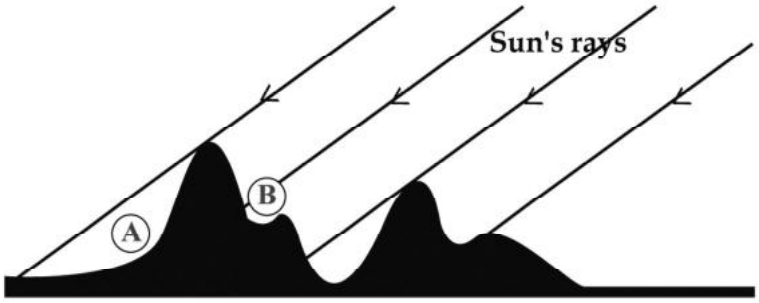


18. Kerala generally experiences a moderate temperature. Justify the statement.

19. Ocean currents influence the temperature distribution on the earth's surface. Explain the statement based on the image provided below.



20. Observe the image



On which of the mountain slopes marked A and B would the sun's light be received more? Explain how landform influences temperature distribution.

21. The temperature is not the same everywhere and at all times on Earth. Spatio-temporal variations are experienced in temperature distribution. Explain how the following indicators influence temperature distribution:

* Latitudinal location
* Altitude from sea level
* Differential heating of land and sea

22. Which of the following factors influence atmospheric pressure?

a) Atmospheric temperature
b) Friction
c) Pressure gradient force
d) Altitude from sea level

i) a and d influence

ii) b and c influence

iii) a, b, and c influence

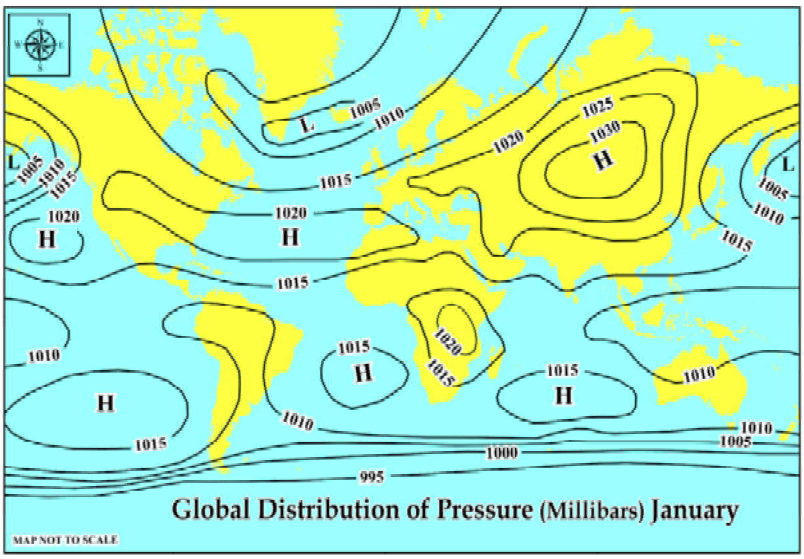
iv) a, b, and d influence

23. Identify the factors that influence atmospheric pressure and explain how each of them influences atmospheric pressure.

24. Why do we experience discomforts like ear blockage when traveling to high-altitude regions like Ooty, Munnar, and Kodaikanal?

25. Atmospheric pressure is lower in coastal areas compared to inland areas. What is the reason?

26. The image provided below is a picture of the Global Pressure Distribution. What do H and L marked in the image indicate?



Global Distribution of Pressure (Millibars) January

27. Read the given statement and reason and select the correct answer.

Statement : The lowest atmospheric pressure on the earth's surface will be in the equatorial regions, and the highest will be in the polar regions.

Reason : Temperature is inversely proportional to pressure.

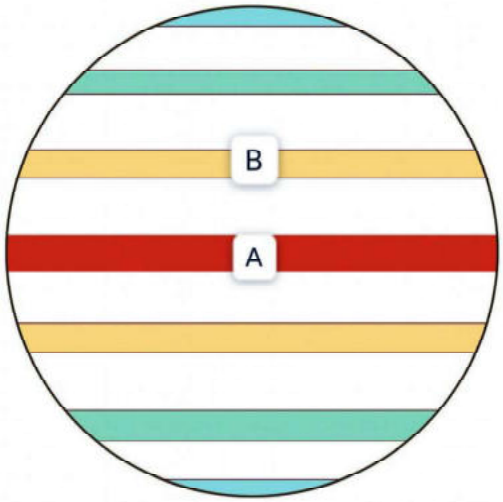
a) Statement correct, reason incorrect

b) Statement incorrect, reason correct

c) Both statement and reason are correct

d) Both statement and reason are incorrect

28. The image provided below is of the Global Pressure Belts. Identify the pressure belts marked A and B in the image and explain the reason for the formation of those pressure belts.



29. The following are atmospheric air movements. Identify the air movement:

a) Vertical movement of air.

b) Horizontal movement of air from high-pressure areas to low-pressure areas.

30. Read the given statement and influence and select the correct answer.

Statement : Coriolis force is the main factor influencing the direction of wind.

Influence : Due to the Coriolis effect, winds in the Northern Hemisphere deviate to the left of their direction of travel, and in the Southern Hemisphere, they deviate to the right of their direction of travel.

a) Statement correct, influence incorrect

b) Statement incorrect, influence correct

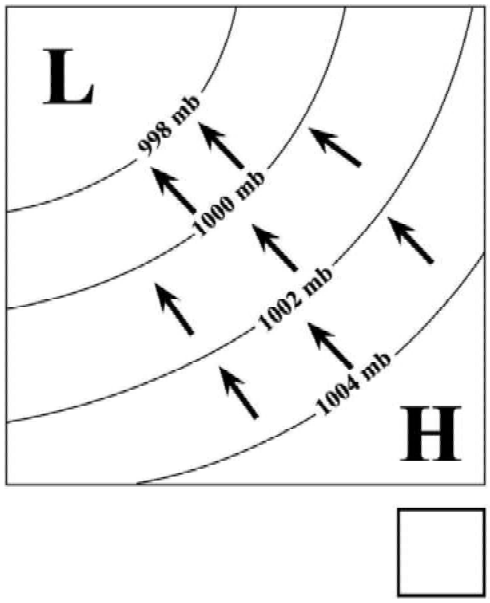
c) Both statement and influence are correct

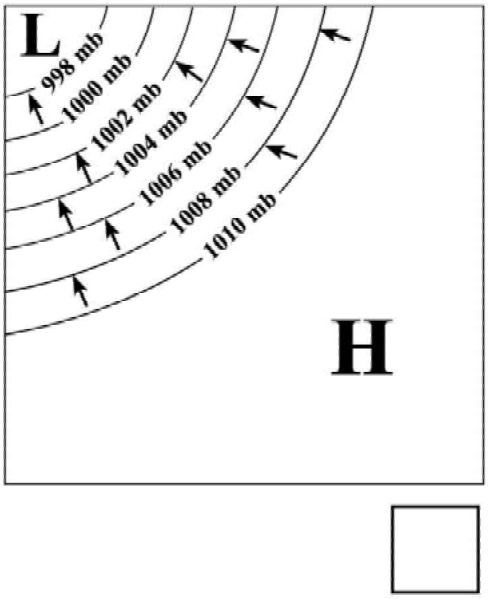
d) Both statement and influence are incorrect

31. Analyse the isobars in the image provided below and find out where the wind speed is greater (A or B), and explain the reason for it.

Or

Analyze the Pressure Gradient Force with the help of the image.





32. Wind is stronger over the ocean surface compared to continents. Justify the statement.

33. What are the factors that influence the direction and speed of wind? Explain.

34. Find the answers to the questions given below:

a) Instrument used to measure wind speed.

b) Instrument used to indicate the direction of wind.

35. Trade Winds blow as North-East winds in the Northern Hemisphere and South-East winds in the Southern Hemisphere. Explain the reason.

36. Read the given statement and reason, then choose and write the correct answer.

Statement: : Compared to the Northern Hemisphere, the westerly winds are stronger in the Southern Hemisphere.

Reason : The frictional force is less because the Southern Hemisphere has fewer land areas and more oceanic areas.

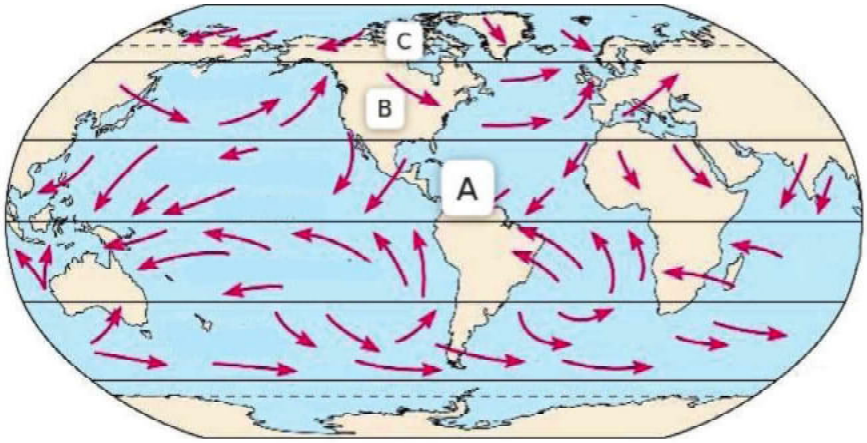
a) The statement is correct, but the reason is wrong

b) The statement is wrong, but the reason is correct

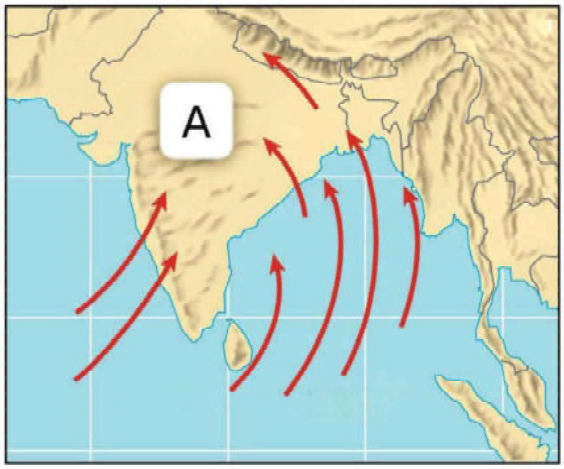
c) Both the statement and the reason are correct

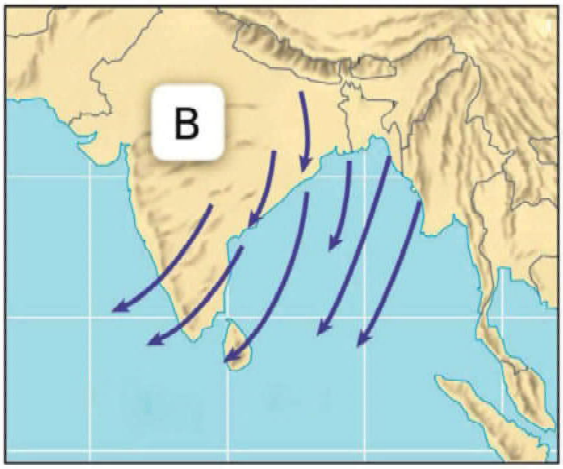
d) Both the statement and the reason are wrong

37. A diagram related to the permanent winds is given. Identify the permanent winds marked as A, B, and C in the diagram and explain between which pressure zones each of them blows.



38. Observe the given A and B diagrams, identify the seasonal winds, and explain them.





39. Read the following statements carefully and find the correct answer.

a) The trade winds are the winds that blow from the subtropical high-pressure belts towards the equatorial low-pressure belt.

b) The westerlies are the winds that blow from the subtropical high-pressure belts towards the subpolar low-pressure belts.

c) The polar winds are the winds that blow from the polar high-pressure belts towards the subpolar low-pressure belts.

a. Statements a, b, and c are correct

b. Statements a and b are correct

c. Statements b and c are correct

d. Statements a and c are correct

40. Examine the given table and identify the correct sequence.

No.	Description	Wind Type
i	During the daytime, wind blows from the sea to the land	a. Valley breeze
ii	During the day, wind blows from the valleys toward the mountain tops	b. Mountain breeze
iii	During the night, air from the mountain slopes moves downward	c. Sea breeze
iv	During the night, wind blows from he land to the sea	d. Land breeze

Choose the correct sequence:

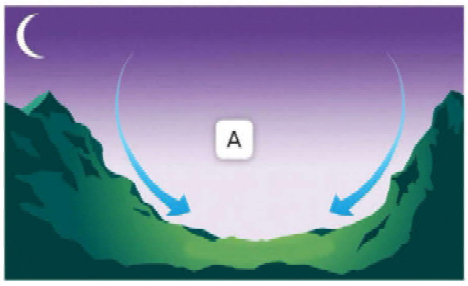
A. i-a, ii-b, iii-c, iv-d

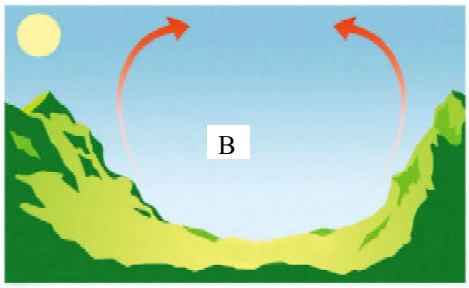
B. i-c, ii-a, iii-b, iv-d

C. i-b, ii-a, iii-d, iv-c

D. i-d, ii-c, iii-a, iv-b

41. Observe the picture A and B given below, identify and explain which are the seasonal winds?





42. Given below are statements related to local winds. Find and write the local winds match the statements.

a) Hot wind blowing across the North Indian Plains.

b) A wind that bring relief from the intense heat blowing in to the Sahara Desert in Africa.

c) Dry hot winds blowing slopes of Alps mountain in Europe

d) Dry hot wind blowing in slopes of Rocky mountains in North America.

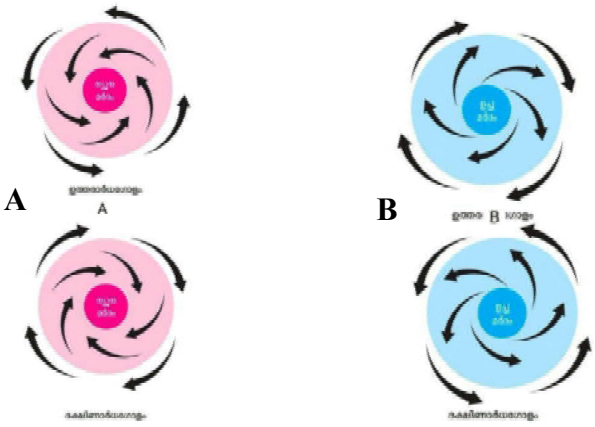
43. A wind that lasts only for a short duration and which the intensity or direction can not be predicted -

a) Variable wind
b) Seasonal wind
c) Trade wind
d) Local wind

44. Which of the following is not a cyclone -

a) Tornadoes
b) Willy willies
c) Typhoons
d) Chinook

45. Two pictures related to winds are given below. Observe the pictures and identify which A and B are variable winds and explain?



46. The following are statements related to cyclones. Identify the incorrect statement.

i) A cyclone is a condition in which strong winds spiral inward from the surrounding areas toward a low-pressure center.

ii) Compared to tropical cyclones, temperate cyclones have a larger extent but are not destructive.

iii) In the Northern Hemisphere, the air circulation in cyclones is clockwise, and in the Southern Hemisphere, it is anticlockwise.

a) iii only is wrong
b) i and ii are wrong
c) ii only is wrong
d) i and iii are wrong

47. Explain the following concepts:

a) Humidity

b) Absolute Humidity

c) Relative Humidity

48. Find the incorrect statement among the following:

i) The condition in which the atmosphere becomes completely saturated with water vapor is called the saturated state.

ii) The critical temperature at which the atmosphere becomes saturated with water vapor is called the saturation point (dew point).

iii) The actual amount of water vapor present in a given volume of air is called relative humidity.

a) i and ii are wrong
b) iii only is wrong
c) ii and iii are wrong
d) ii only is wrong

49. Which of the following is not a form of condensation?

a) Hailfall
b) Fog
c) Clouds
d) Dew

50. We can see different types of clouds in the sky. Read the following statements and identify the clouds.

a) Clouds that appear as thin, feathery patches at very high altitudes in clear weather.

b) Clouds that form as thick layers at low altitudes.

c) Clouds that resemble cotton puffs, formed as a result of convection currents.

d) Dark rain clouds formed at low altitudes.

51. When Kerala receives heavy rainfall from the southwest monsoon, the western parts of the neighboring state Tamil Nadu receive very little rainfall. Analyze the reason.

52. Which form of precipitation are you most familiar with?

a) Dew b) Rain

c) Frost d) Snow

53. In equatorial climatic regions, convectional rainfall occurs almost every day. Explain the reason.

54. Which of the following are forms of condensation?

i) Hailfall ii) Dew

iii) Snow iv) Snowfall

a) i and ii are forms of condensation

b) iii and iv are forms of condensation

c) ii and iii are forms of condensation

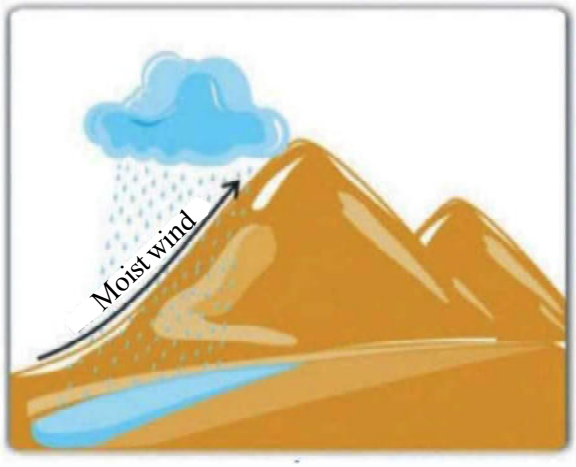
d) ii and iv are forms of condensation

55. The wind that blows steadily in a fixed direction throughout the year is:

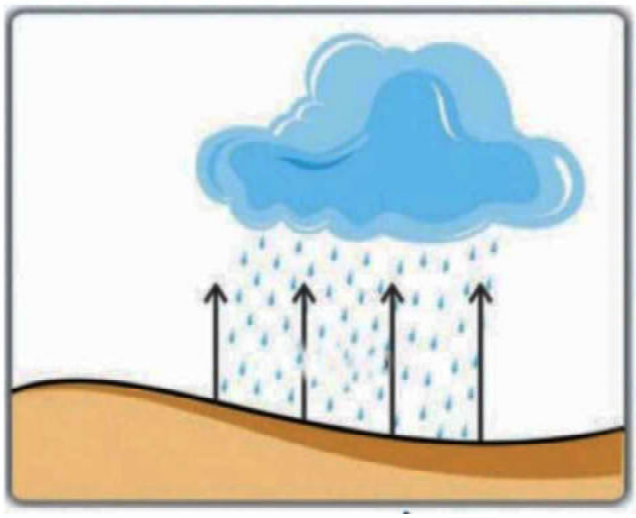
a) Seasonal winds b) Local winds

c) Permanent winds d) Irregular winds

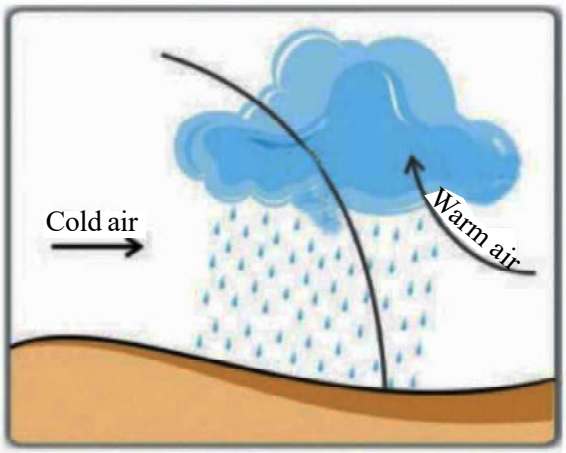
56. Observe the picture and identify which type of rainfall it is. Explain it.



57. Observe the picture and identify which type of rainfall it is. Explain it.



58. Observe the picture and identify which type of rainfall it is. Explain it.



59. What is condensation? What are the different forms of condensation? Explain.

60. What is precipitation? What are the different forms of precipitation? Explain.

61. Due to the low-pressure area formed over the Indian subcontinent during summer, which winds blow from the Indian Ocean?

a) Westerlies b) Cyclones

c) Southwest Monsoon d) Northeast Monsoon

62. In which pressure belt is the doldrum region found?

a) Polar high-pressure belt b) Subtropical high-pressure belt

c) Subpolar low-pressure belt d) Equatorial low-pressure belt

63. Which instrument is used to measure atmospheric pressure?

a) Thermometer b) Hygrometer

c) Barometer d) Anemometer

64. Warm ocean currents and cold ocean currents play an important role in regulating atmospheric temperature. Explain with examples.

Chapter 1

WEATHER AND CLIMATE

ANSWERS

1. c. The short-term atmospheric condition experienced in a particular region is called weather.

2. Climate creates diversity in the plant and animal life and in human existence on Earth. The influence of climatic factors is evident not only in food habits, clothing, housing, and occupations, but also in our physical and mental conditions, and in the variations found among human races. All over the world, agriculture is mainly dependent on climate. From the early sea voyages that changed the course of world history to modern transport and communication systems, everything depends on weather conditions and information.

3. The short-term conditions of temperature, atmospheric pressure, winds, humidity, and precipitation experienced in a region are called daily atmospheric conditions (weather).

   The average of weather conditions experienced over a long period in a large area is called climate.

4. c. ii, iv True

5. Solar Radiation: The enormous energy produced in the Sun through nuclear fusion is radiated in the form of short waves.

   Insolation: The amount of solar radiation that reaches the Earth's surface.

   Terrestrial Radiation: The process by which the Earth's surface re-emits energy in the form of long waves.

   Heat Budget: The nearly complete balance between the total amount of energy the Earth receives daily and the amount it re-radiates back into space.

The heat re-radiated as long waves helps to maintain the Earth's surface temperature in balance — preventing it from increasing or decreasing excessively.
This process of thermal balance is called the Earth's Heat Budget.

6. Conduction: Heat is transferred from the warm Earth's surface to the lower layers of the atmosphere in direct contact with it.

Convection: The air near the Earth's surface gets heated, expands, and rises upward, spreading heat to higher layers of the atmosphere.
Advection: Heat is transferred horizontally through the movement of air (winds).
Radiation: As the Earth's surface becomes heated, it emits energy outward in the form of long waves.

7. b,i, ii, false

8. No, the temperature of the Earth's surface gradually increases from sunrise due to the incoming solar radiation and reaches its maximum by noon. Since the atmosphere gets heated mainly through various heat transfer processes from the Earth's surface, it takes more time for the atmosphere to warm up than for the surface itself. That is why meteorologists record the temperature at around 2 p.m. as the highest temperature of the day.

9. iii. a, c true

10. The difference between the day's highest and lowest temperatures is called the daily range of temperature.

Daily Range of Temperature = Maximum Temperature – Minimum Temperature
The average temperature of a day is called the daily mean temperature.

$$\text{Daily Mean Temperature} = \frac{\text{Maximum Temperature} + \text{Minimum Temperature}}{2}$$

11. Daily Range = 34-26 = 8^0 C

Daily Mean Temperature = 34+26/2 = 30^0C

12. The imaginary lines drawn on maps connecting places of equal temperature are called isotherms.

The imaginary lines drawn on maps connecting places of equal atmospheric pressure are called isobars.

13. a) No - this is due to temperature differences and because land heats up and cools down faster than the ocean surface.

b) Because land and sea heat differently

c) In the Southern Hemisphere, there is more ocean and less land, which also affects temperature distribution.

14.

Zone	Latitude	Range	Description
A	Frigid Zone	90^0N to $66½^0$N & 90^0S to $66½^0$S	From the Arctic Circle to the North Pole and from the Antarctic Circle to the South Pole
B	Temperate Zone	$66½^0$N to $23½^0$N & $66½^0$S to $23½^0$S	From the Tropic of Cancer to the Arctic Circle and from the Tropic of Capricorn to the Antarctic Circle
C	Torrid Zone (Tropical Zone)	$23½^0$N to $23½^0$S	Between the Tropic of Cancer and the Tropic of Capricorn

15. a) Temperature decreases with increasing altitude.

b) For every 1 kilometer of ascent, the atmospheric temperature decreases by 6.4°C

c) This is called the normal lapse rate.

16. As altitude increases, atmospheric temperature gradually decreases. For every 1 km rise, the temperature drops by 6.4^0C.

17. B. In coastal regions, winds blowing from sea to land and back again help to maintain

moderate temperatures. As one moves farther away from the sea, this maritime influence decreases, leading to hotter days and cooler nights inland.

18. Kerala generally experiences moderate temperatures because it is located along the seacoast. In coastal areas, winds blowing from the sea to the land and back help maintain a moderate temperature throughout the year.

19. Warm and cold ocean currents influence the atmospheric temperature of the coastal regions they flow past warm currents raise the temperature, while cold currents lower it.

Example: The warm North Atlantic Current provides relief from severe cold to the western parts of Europe, whereas the northeastern coasts of Canada, located at the same latitude, remain extremely cold for months due to the influence of the cold Labrador Current.

20. B. The region that receives the most sunlight.

Depending on the topography, the amount of sunlight received varies from place to place. The slopes facing the Sun receive more solar radiation and therefore have higher temperatures, while the opposite slopes remain cooler.

21. Latitudinal Position: Regions near the Equator, where the Sun's rays fall almost vertically, have higher temperatures.

Since the Earth is spherical, solar rays strike more obliquely as one moves from the Equator toward the poles.

Thus, temperature gradually decreases toward both poles, leading to the formation of different thermal zones.

Altitude above Sea Level: As altitude increases, the atmospheric temperature decreases at a rate of 6.4°C for every 1 kilometer of elevation.

Differential Heating of Land and Sea: Land heats up and cools down faster than the ocean surface.

Therefore, in summer, land areas experience higher temperatures, and in winter, they experience lower temperatures than nearby sea regions.

22. i. Influenced by both a and d.

23. Atmospheric Temperature, Altitude, and Humidity

- When air heats up, it expands, becomes lighter, and rises, forming low-pressure areas.
- As this rising air cools and contracts, it descends, forming high-pressure areas.
- As altitude increases, the density of atmospheric gases decreases, resulting in a gradual reduction of air pressure.
- On average, air pressure decreases by about 1 millibar for every 10 meters of elevation.
- When the moisture content (water vapor) in the air increases, water molecules replace heavier gases like nitrogen and oxygen.
- Because water vapor is lighter, humid air has lower air pressure than dry air.

24. As altitude increases and the density of air decreases, atmospheric pressure gradually decrease.

25. Coastal regions have more water vapor in the atmosphere.

When moisture increases, lighter water molecules take the place of heavier gases, leading to lower air pressure in humid air compared to dry air.

26. H – High-pressure centers

L – Low-pressure centers

27. C – Both the statement and reason are correct.

28. A – Equatorial Low-Pressure Zone:

Due to high temperatures in the equatorial region, the air becomes warm, expands, and rises.

Since the air moves vertically here, winds do not blow horizontally in this zone.
This region is called the Doldrums (Calm Zone).

B – Subtropical High-Pressure Zone (30° North & 30° South)

The air that rises from the equatorial region moves toward the poles in the upper layers of the atmosphere. As it cools gradually, it descends around 30° North and 30° South latitudes, forming high-pressure zones in these regions.

29. A - Convection

B - Wind

30. a) Statement is correct, but the influence is wrong.

31. B

Since the pressure gradient is higher in certain regions, winds are stronger there.
The pressure gradient is the difference in atmospheric pressure measured horizontally between two nearby areas.
If nearby locations have a significant difference in pressure, the pressure gradient is considered high.
If the difference is small, the pressure gradient is low.
Thus, winds blow more strongly where the pressure gradient is greater.

32. The statement is correct.

There is less friction over ocean surfaces compared to land.
Mountains, hills, forests, and man-made structures obstruct smooth airflow on land, while the sea surface offers fewer obstacles, allowing winds to move more freely and with greater speed.

33. Wind direction – Coriolis Force

Wind speed – Pressure Gradient Force, Frictional Force
The Coriolis Force is the force responsible for the deflection of moving objects on the Earth's surface.

Due to this force, winds deflect to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

The pressure gradient refers to the difference in air pressure measured along a horizontal plane.

When nearby places have a large pressure difference, the pressure gradient is high; when the difference is small, it is low.

Hence, winds are stronger where the pressure gradient is higher.

There is less friction over the sea surface.

Mountains, hills, forests, and human-made structures obstruct the smooth flow of wind over land.

Since such obstacles are absent over the sea, winds move faster over oceans.

34. a) Anemometer – used to measure wind speed

b) Wind Vane – used to determine wind direction

35. Coriolis Force:

The Coriolis Force is the force responsible for the deflection of moving objects on the Earth's surface.

Due to this force, winds deflect to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

Because of the Coriolis effect, Trade Winds blow as Northeast Trade Winds in the Northern Hemisphere and Southeast Trade Winds in the Southern Hemisphere.

36. C – Both the statement and the reason are correct.

37. A - Trade Winds

B - Westerlies

C - Polar Winds

Trade Winds	:	Blow from the Subtropical High-Pressure Belts towards the Equatorial Low-Pressure Belt.
Westerlies	:	Blow from the Subtropical High-Pressure Belts towards the Subpolar Low-Pressure Belts.

Polar Winds : Blow from the Polar High-Pressure Belts towards the Subpolar Low-Pressure Belts.

38. A – Southwest Monsoon

B – Northeast Monsoon

The monsoon refers to the seasonal reversal of wind direction.

During summer, the South Asian landmass, especially the Indian subcontinent, becomes intensely heated, forming a strong low-pressure area.

Winds blow from the high-pressure region over the Indian Ocean toward the low-pressure region over land.

Due to the Coriolis effect, these winds blow from the southwest direction.

As they reach land, they cause widespread rainfall across many regions.

This is the Southwest Monsoon.

During winter, the northern landmass of India, including South India, experiences severe cold, resulting in the formation of a high-pressure area over North India.

Due to this, winds blow continuously from the northeast — from land toward the Indian Ocean.

These winds are generally dry and are known as the Northeast Monsoon Winds.

39. d. Statements a and c are correct.

40. B - i–c, ii–a, iii–b, iv–d

41. A – Mountain Wind

B – Valley Wind

At night, the air on mountain slopes cools, becomes dense, and flows downward - this is called a mountain wind.

During the day, the air on the slopes warms up, expands, and rises, causing wind to blow from the valley toward the mountain top - this is called a valley wind.

42. a) Loo

b) Harmattan

c) Mistral

d) Chinook

43. c) Variable winds

44. d) Chinook

45. A – Cyclones

B – Anticyclones

Cyclones are weather systems where winds spiral inward toward low-pressure centers from all directions. In coastal regions, tropical cyclones cause extremely strong winds and heavy rainfall. These tropical cyclones are known by different names around the world:

Hurricanes – in the Atlantic Ocean and Caribbean

Typhoons – in the Pacific Ocean

Willy-willies – in Australia

Tornadoes – in North America

Compared to tropical cyclones, temperate cyclones cover a wider area, though they are less destructive. In the Northern Hemisphere, air circulation in cyclones is anticlockwise, and in the Southern Hemisphere, it is clockwise. In high-pressure centers, winds spiral outward. In the Northern Hemisphere, the winds in an anticyclone move clockwise, and in the Southern Hemisphere, they move anticlockwise.

.46. a) Only statement iii is wrong

47. Water vapor remains invisibly in the atmosphere. The invisible water content present in the atmosphere is called atmospheric humidity.

The actual amount of water present in a given volume of air is called absolute humidity. The ratio between the actual amount of water vapor present in the air at a given temperature and the maximum amount of water the air can hold at that temperature is called relative humidity.

$$\text{Relative Humidity} = \frac{\text{Absolute Humidity}}{\text{Total Moisture Capacity of Air}} \times 100$$

48. b) Only statement iii is wrong

49. a) Hailfall

50. a) Cirrus cloud

    b) Stratus cloud

    c) Cumulus cloud

    d) Nimbus cloud

51. Between Kerala and Tamil Nadu lies the Western Ghats.

    Since Kerala is on the windward side of the mountains, it receives heavy rainfall. Tamil Nadu lies on the leeward side, where dry air descends, so rainfall is very little.
    Thus, while Kerala receives abundant southwest monsoon rains, the western parts of Tamil Nadu fall under rain-shadow regions, receiving very little rain.

52. a) Rain

53. In equatorial climatic regions, sunlight falls directly, causing very high temperatures. As a result, air heats up rapidly and rises, leading to convection. This process causes rainfall almost every day.

54. c) ii, iii are forms of condensation

55. c) Permanent winds

56. Orographic (Relief) Rainfall:

    Moist air from the sea moves toward the land and is forced to rise when it meets a mountain slope. As it rises, the air cools and condensation occurs, forming rain clouds. This type of rainfall is called orographic rainfall. Heavy rain occurs on the windward side of the mountain, while the leeward side remains dry - this dry region is known as a rain-shadow area.

57. Convectional Rainfall:

    During summer afternoons, rainfall occurs due to the process of convection. Such rainfall is called convectional rainfall. Since it usually happens in the afternoon, it is often referred to as “four o’clock rain.”

58. Cyclonic Rainfall:

In a cyclonic system, when warm air and cold air meet, the lighter warm air rises above the cold air, leading to condensation and rainfall. This is called cyclonic rainfall. The boundaries where warm and cold air masses meet are known as fronts, and this type of rain is also called frontal rainfall.

59. Condensation is the process by which water vapor in the atmosphere cools and changes into tiny droplets of water. Dew, frost, fog, and clouds are the main forms of condensation.

Dew:

At night, as the Earth's surface cools, the layer of air in contact with it also cools. The water vapor condenses into water droplets, which settle on grass, leaves, and other cool surfaces.

Frost:

In regions where the temperature drops below 0°C, especially during the night, ice crystals form instead of dew.

Fog:

When the air cools, water vapor condenses into fine droplets that remain suspended in the atmosphere. Fog forms when condensation occurs around dust particles in the lower layers of the atmosphere.

Clouds:

Clouds form when condensation occurs around fine dust particles in the atmosphere. As condensation continues, the size of the water droplets in clouds increases. When they become too heavy to resist gravity, they fall to the Earth in different forms. This process is called precipitation.

60.
- The various forms of precipitation are rain, snowfall, and hail.
- Rain is the most common form of precipitation experienced as water droplets.
- In cold climatic regions or during winter, when the temperature falls below 0°C, precipitation occurs as tiny ice crystals - this is snowfall.

- When the condensed water droplets in clouds repeatedly freeze in different layers of the atmosphere, they form ice pellets that fall to the Earth as hailstones.

61. c) Southwest Monsoon

62. Equatorial Low-Pressure Zone

63. c) Barometer

64. Warm and cold ocean currents cause the atmospheric temperature along the coastal regions they pass through to increase or decrease respectively.

    For example, the warm North Atlantic Current provides relief from severe cold to the western parts of Europe, while the cold Labrador Current keeps the northeastern coasts of Canada, located at the same latitude, in freezing conditions for several months.

Chapter 2

CLIMATE REGIONS AND CLIMATE CHANGE

1. Which of the following statements regarding the monsoon climatic regions are correct? (1)

   i) This climate region extends up to 10^0 North and South of the equator

   ii) Long and humid summer and short dry winter

   iii) Pampas grass lands are found

   iv) Tropical deciduous forests.

   a) i, ii are correct b) ii, iv are correct

   c) iii, iv are correct d) i, iii are correct

2. Which of the following statements are true about Bushmen people? (1)

   i) Atacama desert in South America

   ii) Kalahari desert in Africa

   iii) Thar desert in Asia

   iv) Indigenous tribal communities in hot desert

   a) i, ii are correct b) ii, iv are correct

   b) iii, iv are correct c) i, iii are correct

3. Tall evergreen trees such as Oak and Sequoia, evergreen conifers such as pine and fir, and shrubs are found here. This statement is associated with which climate region? (1)

4. In the tropical region, hot deserts are located mostly on the western margins of continents. What is the reason? (2)

5. Complete the table by including some of the major climatic regions of the world and the peoples who lived here. (3)

Climatic regions	**Tribal communities**
Equatorial climatic region	a) .....
b)	Maasai Tribal people
Hot deserts	

6. Complete the table related to Tundra region. (3)

Location	a) .....
Winter temperature	b) .....
Indigenous semi nomads	c) .....

7. Along with the natural climatic changes, human interventions also causes changes in the world climate. Classify and List out the following into natural and anthro pogenic.

(Oil mining, Deforestation, Volcanic eruption,
Industrialization, Ocean currents, Changes in the land use) (3)

8. Write any three unique features of the Mediterranean Climatic region? (3)

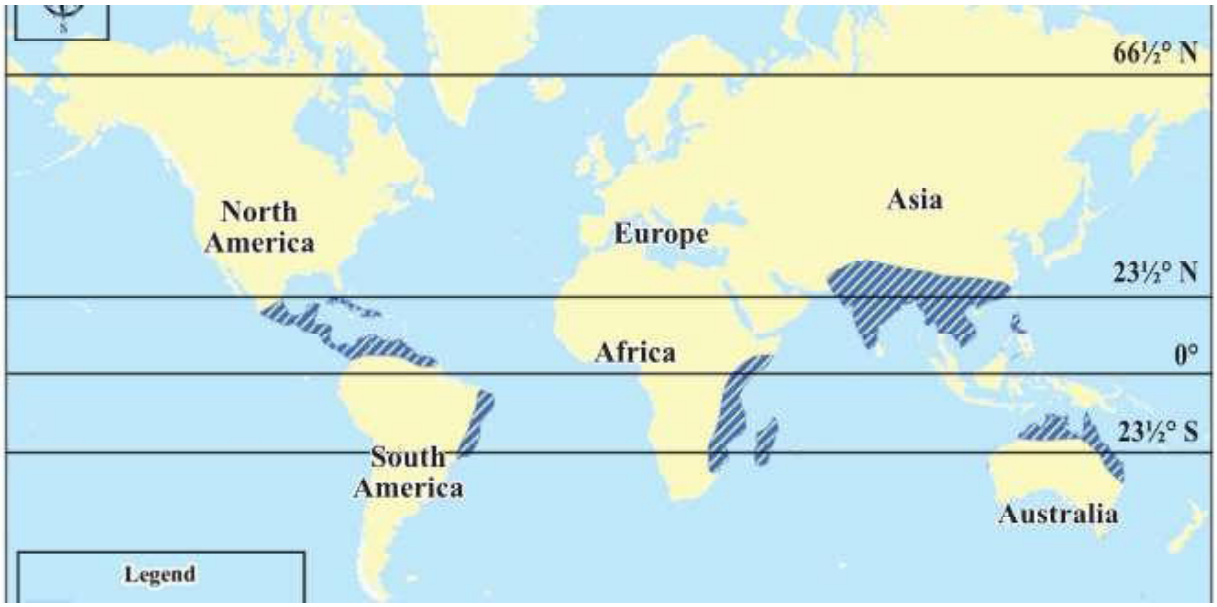
9. Complete the table below the temperature grassland. (4)

Continent	**Name of the grass land**
a)	Pampas
Asia	b) .....
c)	Prairies
Australia	d) .....

10. Prepare a note on Equatorial climatic region based on the hints given below. (4)

- Latitudinal Extension
- Unique Climate

11. Observe the map below and answer the questions. (4)



a) Which climatic region is shaded?

b) What are the characteristics of the climate in this region?

12. Prepare a note on comparing the lifestyles of people and their means of livelihood in the savanna climatic region and hot desert? (4)

13. Mark and Name the grasslands of the savanna climatic region on an outline map of the world. (4)

14. Mark the following hot deserts on the world map. (4)

a) Atacama b) Sahara

c) Namib d) Australian desert

15. What are the features of the climate in the Temperate grass land? (4)

16. Mark the location of the following Temperate grass lands on the given world map. (4)

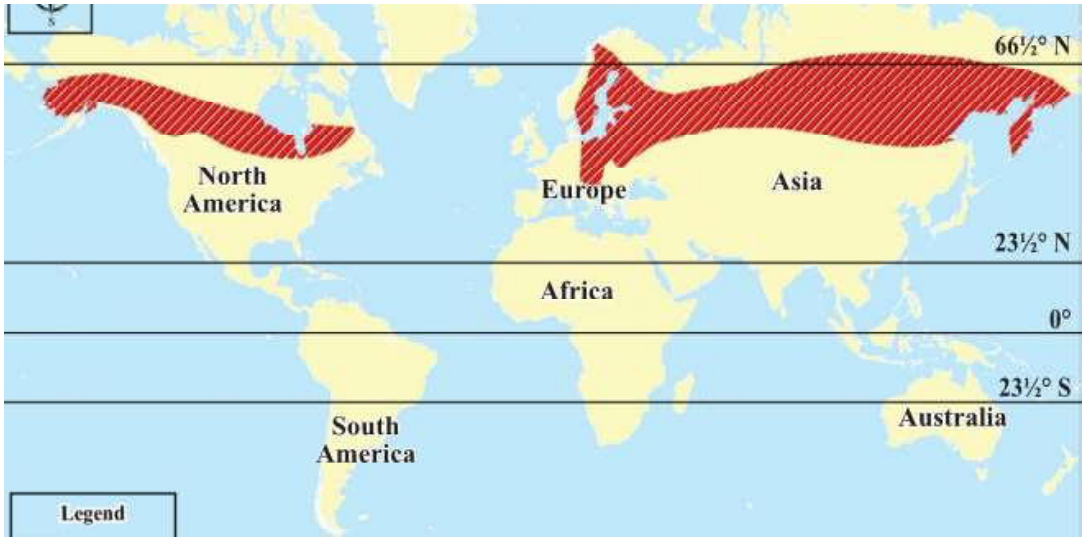
a) Prairies b) Steppes c) Pampas d) Veld e) Downs

17. How does climate change affect human life? (4)

18. International initiatives to protect climate and environment have begun. Some of such initiatives are given below. Complete the table by adding the missing ones. (4)

International Initiatives	**Year**	**Place**	**Interventions**
World meterological Organisation	1950	a)	Organises world climate conferences
Earth summit	b)	Rio de Janeiro	Prepared UN Agenda 21
c)	1997	Kyoto	Reduce the amount of green house gases in the atmosphere
G20 Summit	2023	New Delhi	d) .....

19. Answer the questions related to the climatic region on the map below (5)



a) Which climatic region is marked on the map?

b) Between which latitudes is this region located?

c) What is the temperature and annual rainfall here?

d) Characteristics of flora

e) What are the main economic activities here?

20. What are green house gases? Explain the activities that release green house gases into the atmosphere and the consequences of the gradual increases in green house gases? (5)

Chapter 2

CLIMATE REGIONS AND CLIMATE CHANGE

ANSWERS

1. b) ii, iv are correct
2. b) ii, iv are correct
3. Mediterranean climatic region
4. The trade winds travel across the continents and reach the western margins, the wind loss its moisture and become dry. Therefore the western margins of continents remain dry throughout the year.
5. a) Pygmy
   b) Savanna climatic region
   c) Bushmen
6. a) North of the Arctic circle in Alaska, Canada, Greenland
   b) -25^0C to -40^0C
   c) Eskimo, Lapas
7. Natural causes - Volcanic eruption, Ocean currents
   Anthropogenic - Oil mining, Deforestation, Industrialization, changes in land use
8. It is the region that experiences dry summer and humid winter
   Temperature of around 20-250^0 C is experienced in summer.
   Highest temperature during winter is 10^0C to 16^0C
   Winter rainfall of 30 to 75cm
9. a) South America
   b) Steppes

c) North America

d) Downs

10.
- This climatic region extends up to 10^0 North and South of the equator
- High temperature and high rainfall throughout the year
- Sun's rays fall almost vertically throughout the year
- High temperature causes higher air convection and convectional precipitation
- These areas receive rainfall everyday in the afternoon

11. a) Monsoon climatic region

b)
- Monsoon climate is characterised by long and humid summer and short dry winter.
- Diurnal range of temperature is very low in coastal areas and very high in the interiors.
- 50cm of rainfall to areas receiving over 1000cm of annua rainfall can be found in this region.

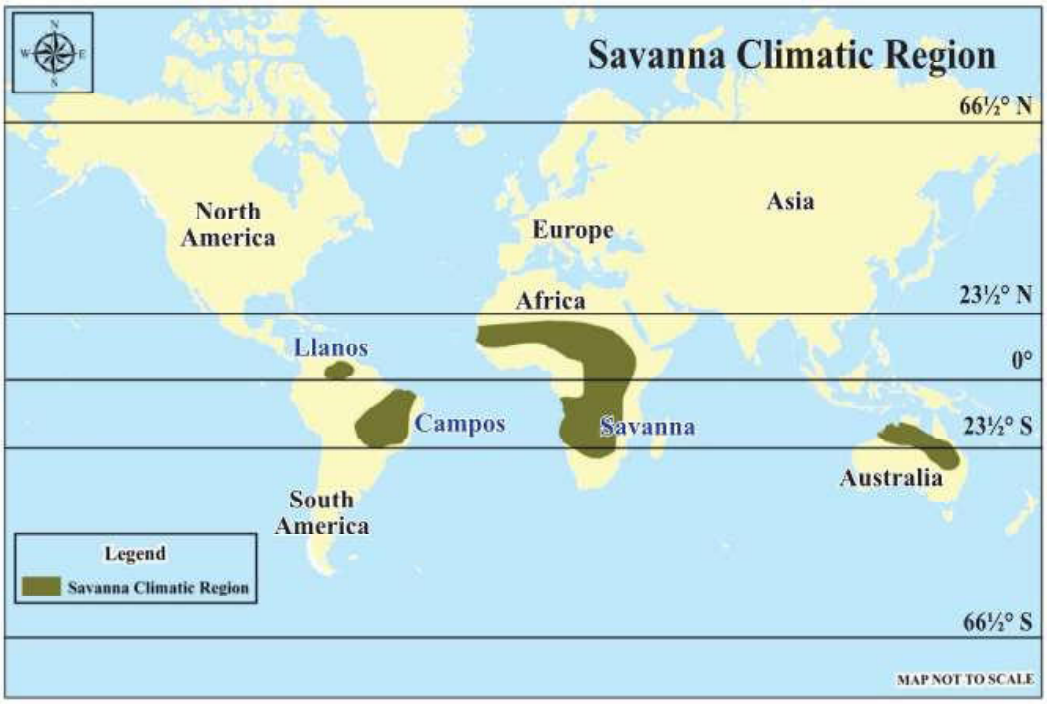
12. **Savanna Climatic Region**

Massai, an indigenous tribe of the African Savanna. Animal husbandry and agriculture are the means of livelihoo of the people. Cash crops are cultivated extensively in the savanna areas of former European colonies.

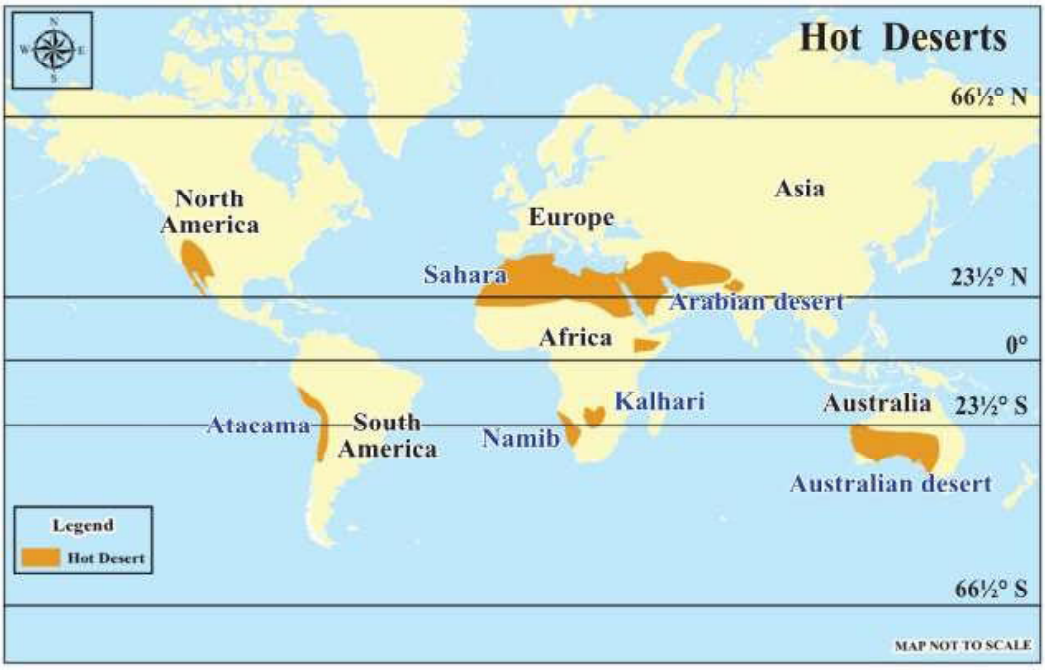
Hot desert

These areas are sparsely populated due to the unfavourable climate and other factors. However there are indigenous tribal communities, who have adapted to the adverse conditions in most of the desert regions. Bushmen of Kalahari Desert are an example for this. Agriculture and animal husbandry are the main means of livelihood in deserts.

13.

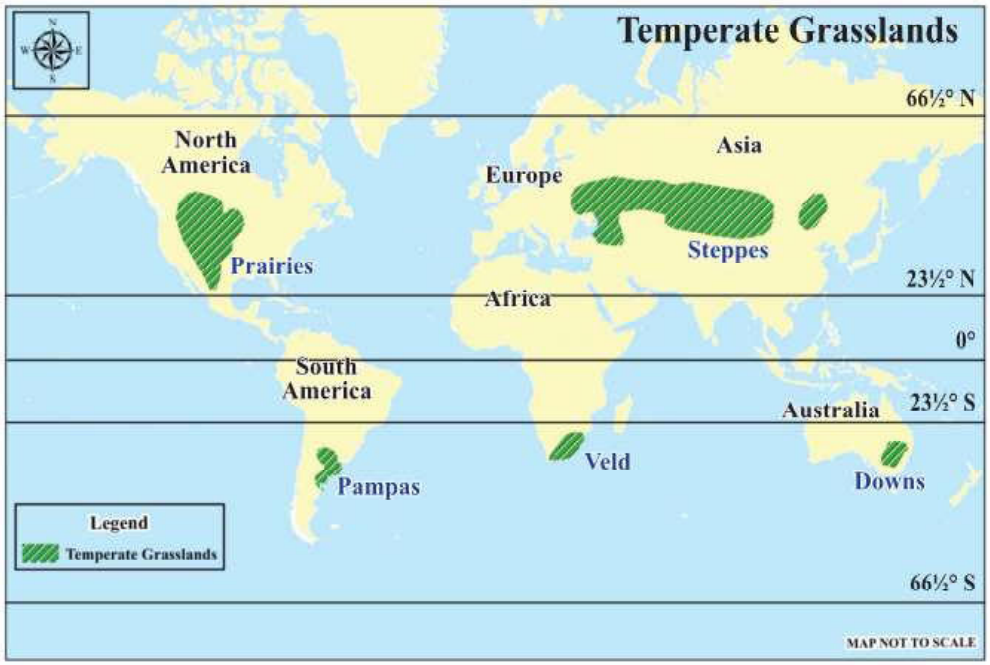


14.



15.
- Short summers and long winters are the characteristics of the temperate grass land.
- These regions experience high temperature in summer
- Average winter temperature ranges from 20^0 to 13^0C
- Rainfall here ranges from 25cm to 60cm.

16.



17.
- Global warming
- Glaciers are melting massively
- Rising the sea level
- Changes in rainfall
- Agriculture crop failure
- Water shortage
- Natural disasters

18. a) Geneva
    b) 1992
    c) Kyoto protocol
    d) One earth, one family, one future.
    Green development, climate, finance,
    overal development

19. a) Taiga region
    b) This region located between latitudes of 55^0 and 70^0 in the northern hemisphere.
    c) Short summers and long winters are experienced there.
    Summer temperature is from 150^0C to 20^0C while winter temperature drops up to

13^0C - 25^0C. This region receives an annual rainfal of 50cm to 70cm. In winter, precipitation is in the form of snowfall.

20. Certain gases in the atmosphere are capable of trapping the solar energy in the atmosphere. Such gases like carbon dioxide and nitrous oxide are known as green house gases. Burning of fossil fuels such as coal and petroleum, industrial effluents and solid waste are the source of excess green house gases in the atmosphere. Globa warning accelerates climate changes. It causes massive melting of glaciers and gradual rise in sea level.

•

Chapter 3

FROM THE RAINY FOREST TO THE LAND OF PARMAFROST

1 Mark questions

1. Which of the following countries that are included in the equatorial climate region?
   a) Indonesi, Nigeria, Brazil, Malaysia
   b) Brazil, Alaska, Mongolia, Columbia
   c) Nigeria, Algeria, Kazakhstan, Equidor
   d) Malaysia, India, Australia, Algeria

2. Statement 1 : Most of the wild life thrive in trees.
   Statement 2 : The animals which spend most of their lives in trees are called arboreal animals .
   a) Statement one is correct and 2 is incorrect.
   b) Statement 1 and 2 are incorrect
   c) Statement 1 and 2 are correct
   d) Statement 1 is incorrect and 2 is correct

3. The strong snow storms that blow over tundra region?
   a) Chinook b) Blizzard
   c) Harmattan d) Foehn

4. Secondary forest spring up in areas used after shifting cultivation. The name of such a secondary forest in Malaysia?
   a) Selvas b) Lianas
   c) Lalang d) Belukar

5. Which are the correct location that belong to Tundra region?
   a) Nepal, Siberia, Greenland
   b) Northern Scandinavia, Alaska, Bhutan

c) Siberia, Alaska, Iceland

d) Greenland, Northern Canada, Mexico

6. Statement 1 : Horizontal movement of air is minimal in this region . It is also where the trade winds from both hemispheres converge.

   Reason : The equatorial region receives a high rate of insolation throughout the year. As a result, a low - pressure region develops along the equator.

   a) Statement is correct and reason is incorrect

   b) Both the statement and reason are correct

   c) Statement is incorrect and reason is correct

   d) Both the statement and reason are incorrect

7. Find the right one

A	B
1. Africa	a) Lapps
2. Malaysia	b) Pygmies
3. Siberia	c) Orang Asli
4. Scandinavia	d) Samoyeds

   a) 1c, 2b, 3a, 4d

   b) 1b, 2c, 3a, 4d

   c) 1d, 2c, 3b, 4a

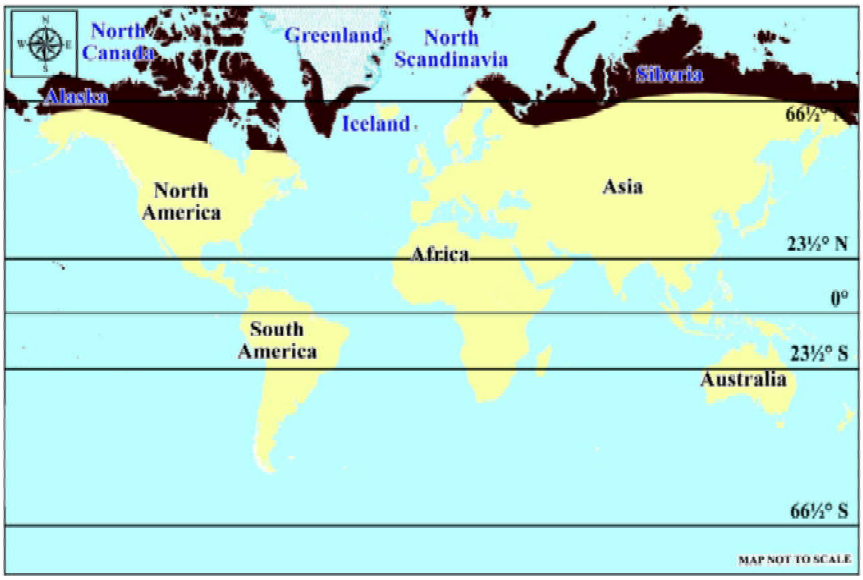
   d) 1b, 2c, 3d, 4a

8. In which climatic zone is Kilimanjaro, the highest mountain peak in Africa situated?

9. What is the name of the rainforest found in the Amazon Basin?

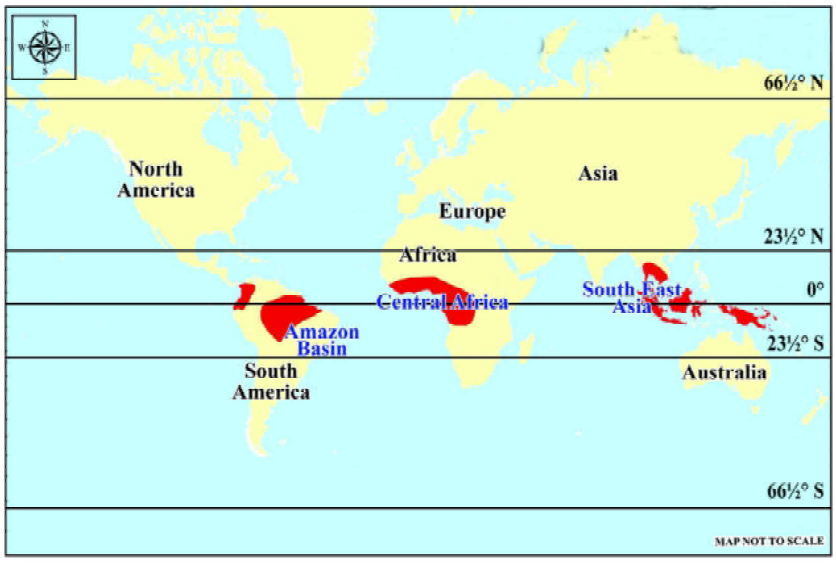
10. Wildlife in this region includes lemurs, chimpanzees, orangutans, tree dwelling reptiles, hippopotamuses, alligators and many birds such as parrots toucan and hornbills. Which climatic zone is this statement about?

11. Examine the map given below and write down the climate regions mark on the map?



2 Mark questions

1. Equatorial climatic region does not experience winter. Why this peculiar type of climate is experienced here?
2. What are the reason for the formation of Doldrums in equatorial region?
3. Apart from convection rainfall ,what are the other two types of rainfall that occur in the equatorial region?
4. Why long nights and days occur in the area that lies north of the Arctic Circle?
5. Examine the given map



a) Find out which climatic zone this refers to?

b) Which continents are included in this climatic zone?

6. • The temperature remains Same throughout the year
   • There is no significant variation in the annual and Diurnal ranges of temperature.

   The mean monthly temperature and the mean annual temperature are both around 27^0 Celsius.

   The above is a characteristics of a climatic zone

   a) What climatic zone is this?

   b) Find the other three features of this area?

7. How the climate in the equatorial region differs from the climate of Kerala?

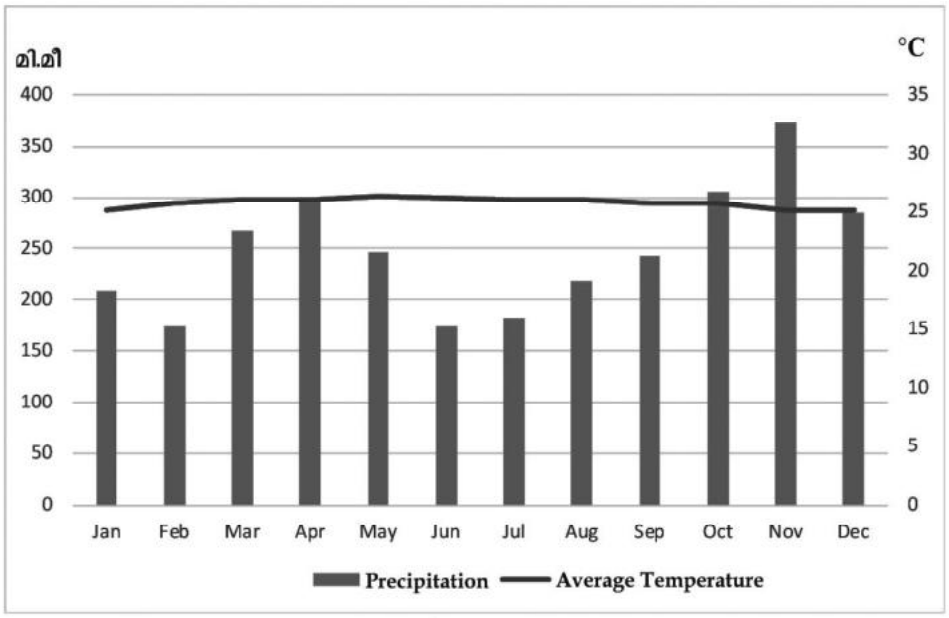
8. What are the reasons for the equatorial evergreen forest are being called the lungs of the world?

9 Herbivores animals are not found in the rainforest of the equatorial climatic zone. And there are very few carnivores animals that prey on them.Why?

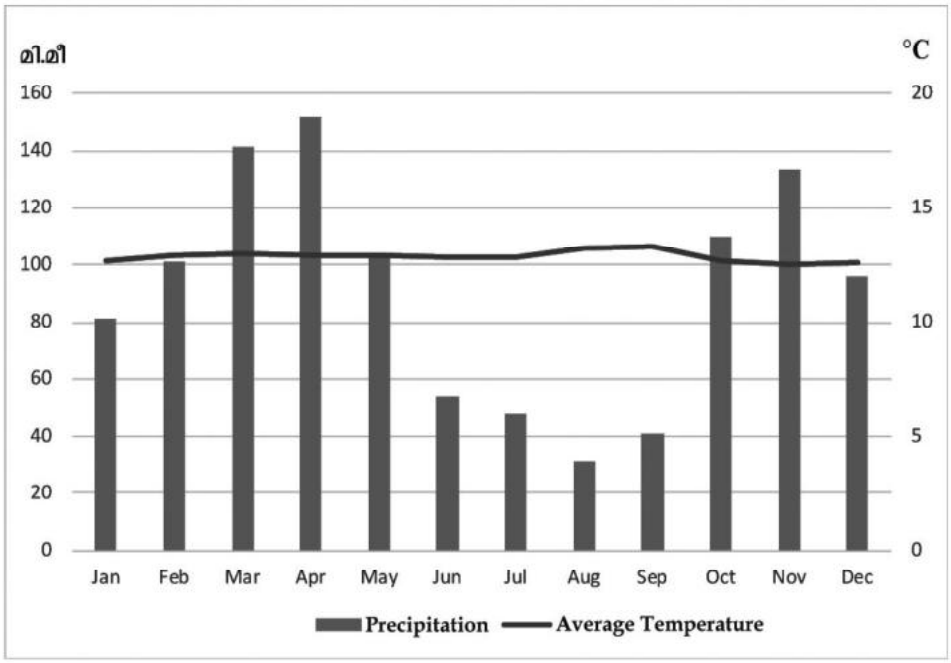
10. The Tundra region can be categorized into Arctic Tundra and Alpine Tundra. Find out the regions to which each type of Tundra belongs.

3 Mark questions

1. The figures given below show the distribution of annual rainfall and temperature recorded at two different locations in the equatorial climatic zone. Analyse the figures and find the answers to the questions given below.



Kuala Lampur



Bogota

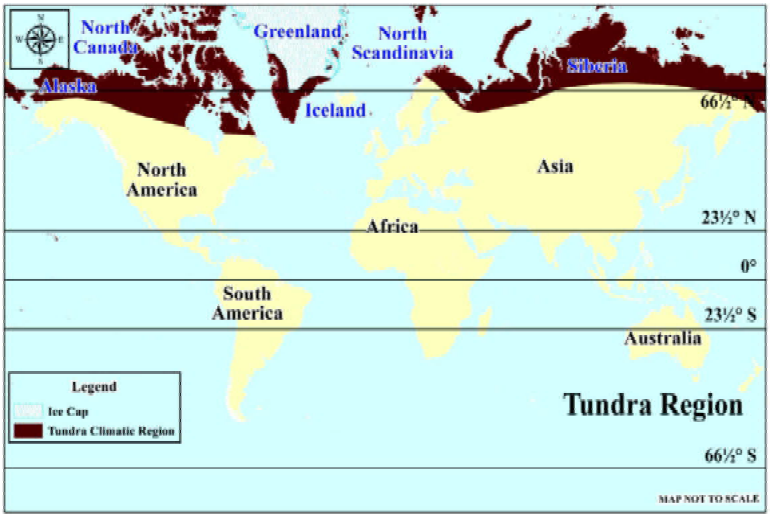
a) Find the highest and the lowest mean monthly temperatures for each place.

b) What is the annual range of temperature at each place?

c) Is there any month that does not receive rainfall?

2. Which are the important native groups found in the equatorial climatic zone?

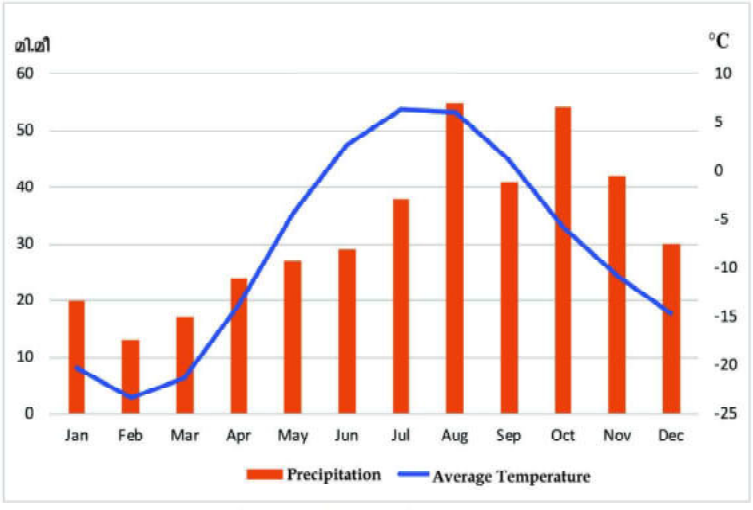
3.



Examine the map and find out which continent that Tundra region extends over and complete the table given below.

Places	Continents
Siberia	
Greenland	
Iceland	
North Scandinavia	
North Canada	
Alaska	

4. Analyse the diagram and answer the questions below



Upernavik in Greenland

a) Month that receives the maximum rainfall

b) Month that receives the minimum rainfall

c) Month in which the maximum average temperature is recorded

d) Month in which the minimum average temperature is recorded

e) The maximum average temperature and the minimum average temperature

f) The annual range of temperature

4 mark questions

1. Analyse the characteristics features of the flora in the equatorial climatic zone?

2. Examine the changes in the lifestyle of the Nomadic tribal people of the Tundra region?

3. What should be included in a seminar paper prepared by Anu on the topic "the challenges faced by the equatorial rainforest"?

4. Write the characteristics features of the natural vegitation in Tundra region?

5 mark questions

1. Examine the features of the climate that prevail over equatorial climate region|

Chapter 3

FROM THE RAINY FOREST TO THE LAND OF PARMAFROST

ANSWERS

1 Mark

1. a) Indonesia, Nigeria, Brazil, Malaysia
2. c) Statement 1 and 2 are correct
3. b) Blizzard
4. d) Beukar
5. c) Siberia, Alaska, Iceand
6. a) Statement is correct and reason is incorrect
7. d) 1b, 2c, 3d, 4a
8. The equatorial climatic region
9. Selvas
10. The equatorial climatic region
11. Tundra Region

2 Mark

1.
   - The equatorial region receives vertical soar rays throughout the year.
   - The high rate of insolation received here causes consistantly high temperature.
   - Because of this the region does not experience Winter.

2.
   - The equatorial region receives a high rate of insolation throughout the year
   - As a result a low pressure region develops along the equator.
   - Horizontal movement of air is minimal in this region.
   - These are the reasons for the formation Doldrums in equatorial region.

3.
   - Orographic rainfall
   - Cyclonic rainfall

4.
- When the sun's apparent position is in the Northern Hemisphere the North Pole experiences day for six months.
- When the sun's apparent position is in the Southern Hemisphere the North Pole experiences night for around six months.

5. a) Equatorial climate region
   b) South America, Africa and Asia

6. a) Equatorial climate region
   b)
   * There is no winter
   * This region receives vertical solar rays throughout the year
   * A high rate of evaporation
   * Heavy downpours of convectional rain in the afternoon

7.
- The climate in the equatorial climate region is generally very hot and humid throughout the year and heavy downpours of convectional rain in the afternoons.
- Kerala has a tropical monsoon climate, which means it experiences a distinct wet and dry season.

8.
- These evergreen rainforests absorb carbon dioxide and produce Oxygen at a massive rate. As a result these forests are often referred to as the 'Lungs of the world'.

9.
- Since sufficient sunlight does not penetrate to the floor in these dense forests, undergrowth is absent.
- As a result herbivores that feed on this undergrowth are not commonly seen.
- Consequently carnivores that prey on them are also negligible in number

10.

Artic Tundra	Alpine Tundra
In parts located to the north of Taiga in Alaska, Northern Canada, Siberia, Greenland, Iceland, Scandinavia	In high mountainous regions

3 Mark

1. a) *Kuala lampur*

Highest mean monthly temperature - 23.3^0C

Lowest mean monthly temperature - 25.1^0C

Bogota

Highest mean monthly temperature - 13.3^0C

Lowest mean monthly temperature - 12.5^0C

b) Kuala lampur

Annual range of temperature - 26.3^0C - 25.10^0C = 1.2^0C

Bogota

Annual range of temperature - 13.3^0C - 12.5^0C = 0.8^0C

c) Kuala lampur

No, It rains every month

Bogota

No, It rains every month

2.
- Pygmies of Africa
- Indian Tribes of Amazon Basin
- Orang Asli of Malasia

3.

Places	Continents
Siberia	Asia
Greenland	North America
Iceland	Europe
North Scandinavia	Europe
North Canada	North America
Alaska	North America

4. a) August - 55 cm

b) February - 13 cm

c) July

d) February

e) Maximum average temperature 6.4
Minimum average temperature - 23.4

f) 29.8

4 mark

1.
- Luxuriant forests called tropical rainforests are one of the Salient Features.
- The rain forest found in Amazon Basin is called Selvas
- There is no particular sea on for seeding, flowering, fruiting and shedding leaves.
- All these process occur year round in the tropical rainforest
- They remain evergreen throughout the year
- A wide variety of evergreen trees including Ebony, Mahagany, Cinchona, rosewood and others are seen in these forests.
- Besides large trees, smaller palms, climbing plants like lianas, epiphytes like orchids, numerous parasitic plants, ferns and grasses like lalang grow here.

2.
- In the last sixty years the way of life of the Eskimos has undergone tremendous changes.
- In Coastal villages, Eskimos live in houses with modern amenities.
- They now speed boats for fishing
- In some areas schools have been established for Eskimo children to enable them to live in the modern life.
- In the Siberian Tundra large farms have been established for rearing fur-bearing animals on a commercial scale.

3.
- Hot and wet equatorial climate is highly supportive of plant growth. It also en courages the spread of insects and pests.
- As gems and bacteria are more easily transmitted through moist air this leads to a widespread occurence of diseases in the region.
- The spread of insects and pests is also harmful to crops.
- Most of the equatorial climate region is devoid of basic amenities.
- It is too difficult and expensive to construct roads and railway lines through dense forests.

- Wild animals, disease spreading insects and poisonous creatures pose a threat to the lives of those engaged construction work.

4.
- The natural Vegetation is scanty in this region
- Trees are normally absent
- Mosses, Lichens, sedges and bushes are commonly found here
- Dwarf Willows and stunted birches with stand the harsh climatic conditions.
- Some hardy grasses grow in the coastal lowlands.

5 mark

1.
- The temperature remains more or less same throughout the year
- There is no significant variation in the annual or diurnal ranges of temperature
- There is no winter in the equatorial climate region
- The significant rise in temperature causes a high rate of evaporation followed by heavy downpours of convectional rain in the afternoons.
- Along wih convectional rainfall Orographic rainfall is also received in certain places
- In the Doldrums occationally lead to intermittent rain fall of cyclonic origin.

Chapter 4

CONSUMER : RIGHTS AND PROTECTION

1 Mark

1. Read the given statement and reason and choose the correct answer and write it.

   Statement (A) : A customer derives satisfaction from using goods and services

   Reason (R) : The primary objective of all economic activities is customer satisfaction

   a) A and R are correct A is the correct explanation

   b) A is true but R is false

   c) Both A and R are true but R is not a correct explanation of A

   d) A True but R is false

2. Read A and B and write the correct choice

A	B
a) Utility	i) The total amount of utility that a person receives when he continuously consumes several units of a particular commodity
b) Cardinal utility theory	ii) We can measure utility by using the unit of you utils
c) Total utility	iii) The change in total utility that occurs when one additional unit of a commodity is consumed
d) Marginal utility	(iv) The satisfaction derived by the consumer through the consumption of goods and services can be quantified by using Cardinal numbers.

A) a-ii, b-iv, c-i, d-iii

B) a-iv, b-i, c-ii, d-iii

C) a-iii, b-ii, c-i, d-iv

D) a-i, b-iii, c-iv, d-ii

2 Marks

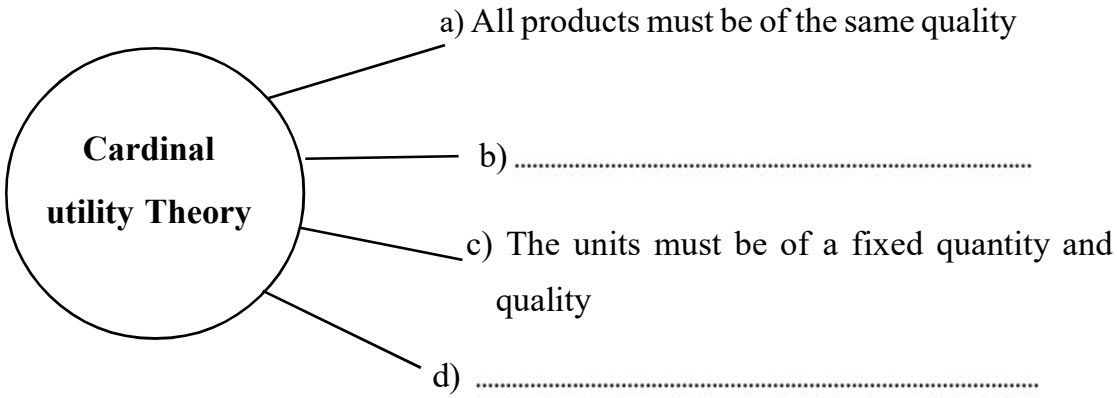
3. Vinu bought a shirt from a shop and used it. One of the factors that influenced his consumption was the attractiveness of the shirt. What other factors could have influenced his consumption? List them.

4. Jinu bought a pair of shoes and used them for the school sports competition and he was satisfied with this which helped him to perform well.

   a) Who was the first economist to attempt to measure happiness?

   b) What is the satisfaction a customer gets from using a product?

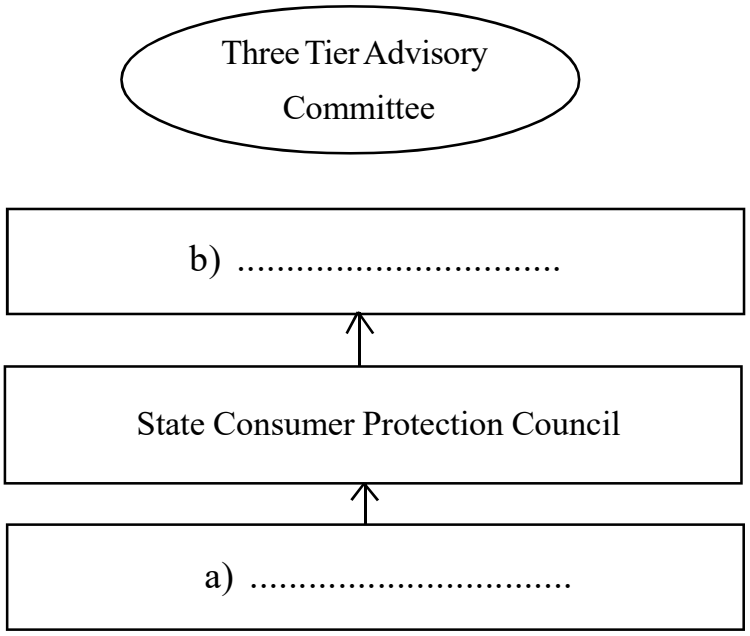
5. Assumptions and limitations of cardinal utility theory are given below. Complete the gaps



6. Users should be able to use it smoothly without any exploits. It needs the help of laws.What are these laws?

7. With the establishment of consumer courts, a three-tier consumer dispute resolution system has been devised for speedy redressal of complaints. Similarly, three-tier advisory committees are also in place.

Fill in the missing parts



8.


പണയ സ്വർണം തട്ടിപ്പ്: ധനകാര്യ സ്ഥാപന ഉടമ അറസ്റ്റിൽ
വീട്ടമ്മയ്ക്ക് ഹോട്ടൽ ഉടമ നഷ്ടപരിഹാരം നൽകണം
ഓൺലൈനിൽ വാങ്ങിയ സ്മാർട്ട് വാച്ചിന്റെ നിറംമാറി; 30,000 രൂപ നഷ്ടപരിഹാരം വിധിച്ച് ഉപഭോക്തൃ കോടതി
Online shoppers beware! Fraud cases on the rise
Beware of catchy advertisements
Customers accuse real estate giant Aliens Group of cheating
Scam Maybe Worth Over ₹300 Cr; Home Minister Promises Action
2.6 ലക്ഷം രൂപയുടെ സൈബർ തട്ടിപ്പിനും ഉപഭോക്താവിന്

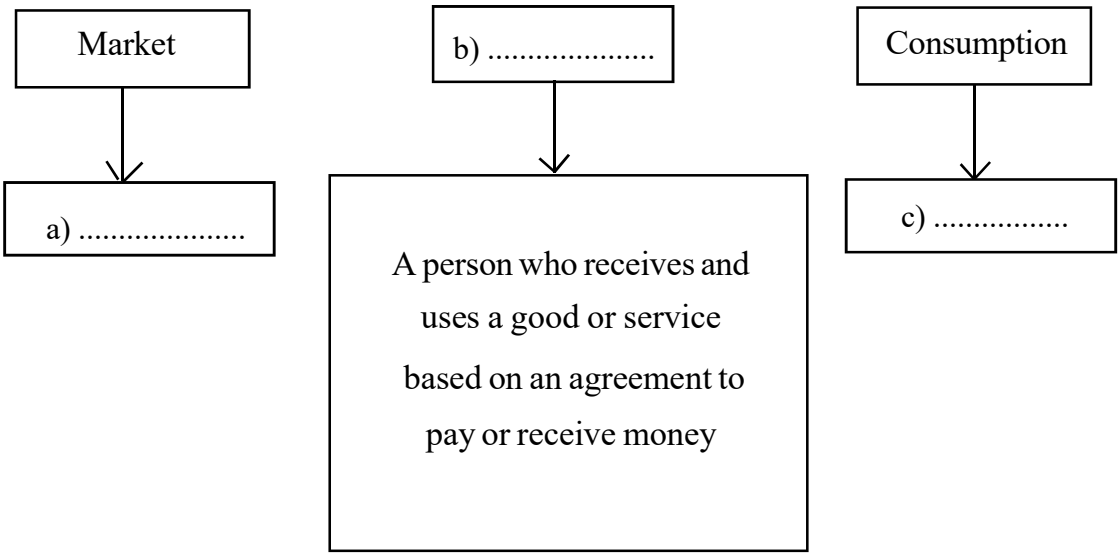
What kind of solutions do you offer to victims of this type of banking fraud after reading the above news story?

9. When a defrauded customer went to court to file a complaint, he found that the shopkeeper had not paid him the bill. So can he file a complaint? What is your opinion?

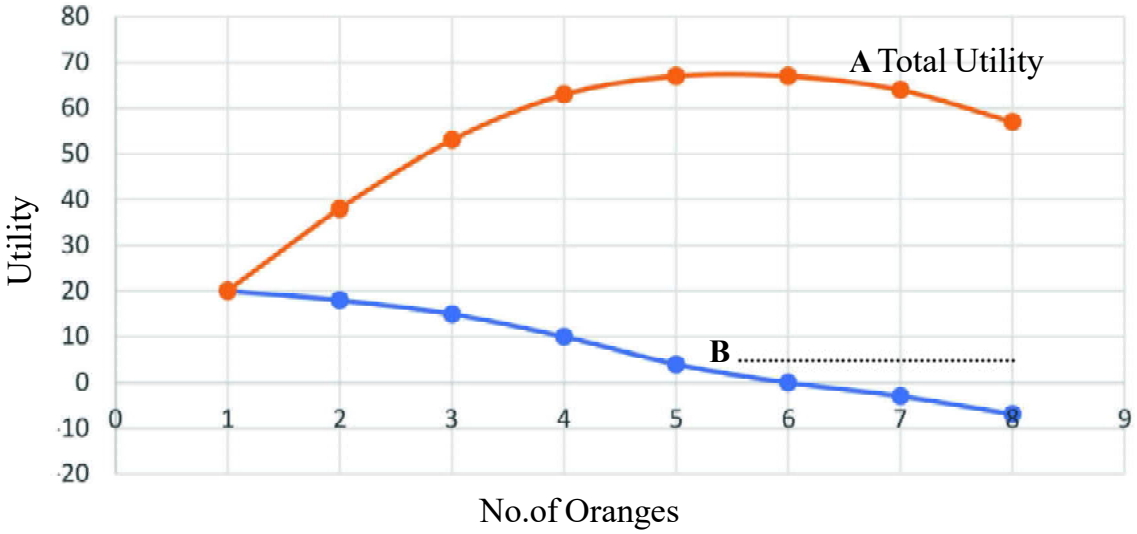
10. FSSAI, AGMARK symbols are printed on the package of a product. What do you understand from products with such symbols?

3 Mark

11. Complete the flowchart below



12.



a) A denotes total utility. But what does B stand for?

b) Write a note indicating the relationship between these documents.

13. Sarath, a class 10 student, went to the market with his parents to buy a new TV for home. What are the things that a consumer should consider while buying a TV? Complete the list.

a) Maximum usable period/ date

b) .....

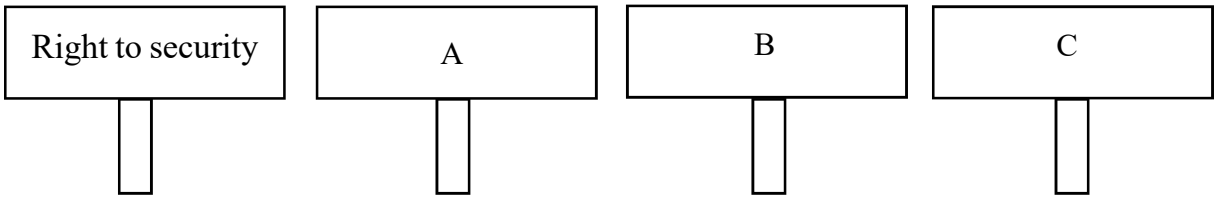
c) .....

d) .....

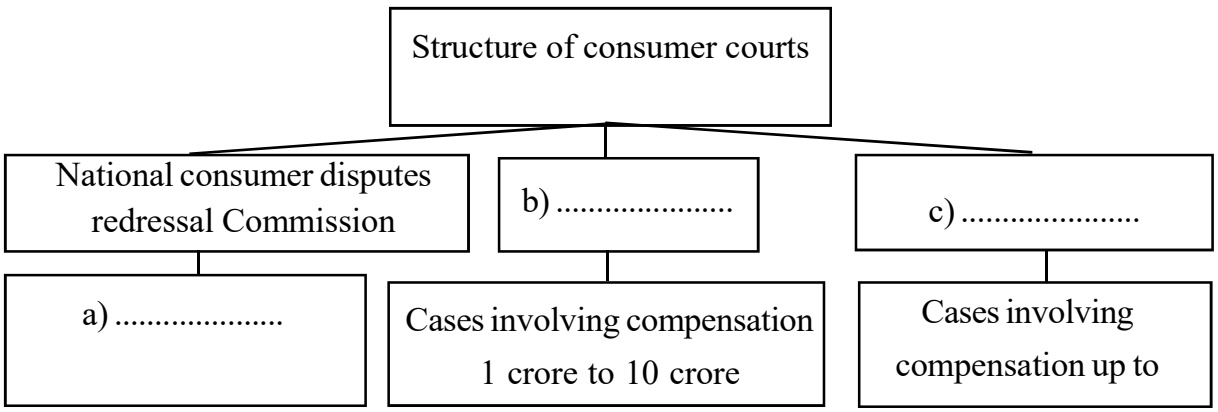
14. Sonu wrote in his class discussion paper that ensuring the right to consumer education is a special feature of the Consumer Protection Act.

What other specifics can Sonu include in his discussion note?

15. Social Science Club has organized a placard making competition to create awareness about consumer rights in the school. Make a placard in the given format.



16. Complete the chart related to formation of Consumer Disputes Redressal Committee



17. Chant, playing the Austin game, has lost a large sum of money from his patron's account. Cheated by Austin, which organizations can Chand approach for compensation?

What are the ways to get compensation?

18. Shyam came abroad through an agency and realised that he had been cheated by being offered a job. Explain the institutions and methods to be approached by Shyam to get rid of fraud.

4 Marks

19. Analyse the table and answer the following questions

Number of Apples	Marginal utility	Total utility(TU)
1	25	25
2	35	10
3	43	8
4	48	5
5	51	3
6	51	0
7	48	-3
8	41	-7

a) What happens to total utility when 1to5 units of apples are consumed?

b) What is the change in total utility after the consumption of the 6 th unit?

c) When ascertain unit of apples in consumed the marginal utility is shown as zero. Which is it?

d) Why does the marginal utility become negative when 7 and 8 unit apples are consumed?

20. Complete the table using the hints provided

Type of goods	Peculiarities	Example
a)	Free goods are those that are abundant in nature and are freely available to every one they need not be paid for	Air
Durable goods	b)	Vehicle
Non durable goods	Non durable goods are items that can be used only for a short period of time	c) .....
Economic goods	d)	Minerals

21. You can find various consumer organizations and individuals in your area working against companies that cheat consumers. Write the objectives of such consumer protection movement

22. What are the institutions and avenues to approach for redress if a person is cheated by health care and educational institutions?

23.



Seeing the above advertisement, Vinu's grandmother asked Vinu what was the purpose of this advertisement. Write down the consumer education objectives given to.

Chapter 4

CONSUMER : RIGHTS AND PROTECTION

Answers

1 Mark

1. a) A and R are true and they are correct explanation of A

2. A. a-ii, b-iv, c-i, d-iii

3. a) Price of goods
   b) Quality of goods

4. a) Alfred Marshall
   b) Utility

5. b) It is assumed that consumer's income, tastes, etc. should be constant
   c) Consumption should be continuous. Also the use of goods should be determined.

6. Consumer Protection Act 1986
   Consumer Protection Act 2019

7. b) National Consumer Protection Council
   a) District consumer protection Council

8. a) Approach the bank grievances redressal mechanism (branch manager)
   b) If the bank does not respond within 30 days you can approach the RBI ombudsman
   c) banking ombudsman toll free number 14448,
   cgmbank@rbi.org.in (Write any two)

9. If a consumer is cheated he can file a complaint with the respective office or consumer courts. Persons with grievances can write a detailed complaint on a white paper and submit it directly to the consumer dispute redressal Court, along with supportingdocuments as evidence. Insisting on bills at the time of purchase assumes significance in this context.

10. FSSAI - Ensures food security

AGMARK this symbol is used to ensure the quality of Agricultural and Forestry products

11. a) Place where sellers and buyers come together to exchange goods and services with each other.
    b) Consumer
    c) Humans buy and use goods and services to fulfill their needs.

12. a) Marginal utility
    b) Total utility is the total amount of utility that a person receive when continuously consumes several units of a particular commodity. The change in total utility that occurs when one additional unit of a commodity is consumed is called marginal utility

13. b) Quality of products
    c) Cost of the item
    d) Warranty/Guarantee
    e) Safety standards

14. • Established the central Consumer Protection authority (CCPA) to protect and enforce the rights of consumers.
    • Covers rules to prevent and fair trade practices by e-governance platforms.
    • Helps in simplifying the consumer dispute resolution process
    • Provides for punishment of those who manufacture or sell adultered counterfeit goods
    • Prohibits the dissemination of misleading advertisements
    • Guarantee's the right to consumer education

    (Write any three)

15. a) Right to safety
    b) Right to choose
    c) Right to seek redressal
    d) Right to consumer education

16. a) Cases involving compensation exceeding rupees 10 crore
    b) District consumer disputes redressal Commission
    c) State consumer disputes redressal Commission

17. a) Complain to Cyber crime cell at cyber crime. government.in
    b) Report to the CERT-IN portal(Indian computer emergency response team)
    c) Submit a complaint through the national consumer helpline or at info @ cert.in org.in

18. a) Can complained Through The Protectorate of immigrants(POE)
    b) A case can be filed under IPC section 420
    c) If you were scammed after arriving at the foreign country you can lodge a complained with the Indian Embassy or High Commission.
    d) Can get in touch with at Pravasi sahayata KendraPBSK toll number 1800-11-3090.

19. a) Total utility increases when 1 to 5 apples are consumed
    b) Total utility decreases after 6th unit
    c) Marginal utility becomes zero when the sixth unit of apple is consumed
    d) Marginal utility becomes negative when 7.8 units of apples are consumed.Because the utility (satisfaction) obtained from it decreases.

20. a) Free goods
    b) Things that last for a long time. They can be reused
    c) Clothes, Shoes
    d) The goods and services that are purchased and used for a price.
    They can often be manufactured or collector from nature.

21. a) To protect the consumer rights
    b) To prevent frauds
    c) To empower consumers
    d) To legislate

e) To ensure the credibility of advertisement

f) To ensure representation of consumers in political forums

22. *Healthcare frauds*

a) File a complaint through the grievance portal of the state/ national /medical council

b) Complain with a report to the director of health

c) File Public Interest Litigation(PIL) in the High Court and Supreme Court for serious health frauds

Frauds related to educational institutions

a) Contact UGC, AICTE, State Board

b) Approach consumer forum

c) Lodge a complaint with the police

d) Contact the ministry of education

23. 1. To protect consumer rights

2. To make consumers aware of their responsibilities

3. To ensure safety in trade

4. To identify ethics and regulations in the market

5. To create awareness about Consumer Protection laws

6. To ensure health safety

7. To promote consumer justice

Chapter 5

MONEY AND ECONOMY

1 Mark

1. What is the universally accepted medium for exchanging goods and services?

   a) Debt b) Deposits

   c) Money d) Interest

2. Statement (A) : Anything accepted in the exchange of goods and services can generally be called money.

   Reason (R) : Exchange of goods and services between individuals and between production units became easier.

   Choose the correct answer from the options given below.

   a) Both (A) and (R) are correct ; and (R) is the correct explanation of (A)

   b) Both (A) and (R) are correct but (R) is the correct explanation of (A)

   c) (A) is correct but (R) is wrong

   d) (A) is wrong but (R) is correct.

3. Statement I : The Reserve bank is the central bank of India.

   Statement II : It was established on April 1935 in Kolkata under the Reserve Bank of India Act 1934.

   Analyse the above statements and choose the correct answer from the options given below.

   a) I and II are correct

   b) Only I is correct

   c) Only II is correct

   d) I and II are not correct

4. Assertion (A) : Reserve bank of India has the power to print and issue all currencies except coins and one rupee notes.

Reason (R) : The coins and one rupee notes are printed and issued by the Ministry of Finance,Government of India.

Choose the correct answer from the options given below.

a) Both (A) and (R) are correct ; and (R) is the correct explanation of (A)

b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

c) (A) is correct but (R) is wrong

d) (A) is wrong but (R) is correct

5. What is the measure taken by the government in 2016 to withdraw 500 and 1000 rupee currency notes from circulation?.

a) Demonetization b) Nationalization of Banks

c) Formation of Banks d) Printing currency

6. Statement I : An increase in the general price level of goods and services is known as inflation.

Statement II : When inflation increases in the economy, the purchasing power of money increases.

Choose the correct answer from the options given below.

a) Only I is correct b) Only II is correct

c) I and II are correct d) I and II are not correct

7. Statement I : Recurring deposits are deposits of a fixed amount of money at regular intervals for a specific period of time.

Statement II : Such deposits earn lower interest rates rather than savings deposits.

Choose the correct answer from the options given below.

a) I and II are correct

b) Only I is correct

c) Only II is correct

d) I and II are not correct

8. Statement I : RTGS is a system introduced by the RBI to transfer large amounts of money.

Statement II : UPI is a payment system developed by the National Payments Corporation of India.

Choose the correct answer from the option below.

a) Only I is correct b) Only II is correct

c) I and II are correct d) I and II are not correct

9. Assertion (A) : Credit Deposit Ratio measures the proportion of a bank's de posits that are used for loans.

Reason (R) : A high credit deposits ratio indicates that banks have lent out a large portion of the deposits they have received.

Choose the correct answer from the options below.

a) Both (A) and (R) are correct ; and (R) is the correct explanation of (A)

b) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

c) (A) is correct but (R) is wrong

d) (A) is wrong but (R) is correct

10. Choose an example of microfinance, founded by Professor Muhammed Yunus in Bangladesh in 1983, from the following examples.

a) Gramin Bank b) Kudumbashree

c) Self help groups d) Neighbourhood Groups

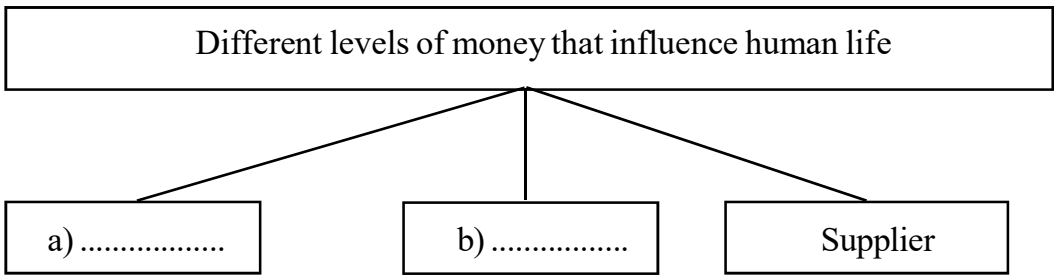
2 Mark

11. A rubber farmer sold the rubber he produced and used the money to buy books for his child's education.

a) Which function of money making this process possible?

b) What is the benefit of convert Rubber into money?

12. Which are the different levels of money influence human life ? Complete the diagram below.



13. Which bank is known as Banker's Bank? Explain Why?

14. Fill in the blanks appropriately according to the example given below.

Example : M1 = Coins and currency notes held by the public and the savings deposits in commercial banks

M2 = M1 + savings deposits in post office savings banks

M3 = a) .....

M4 = b) .....

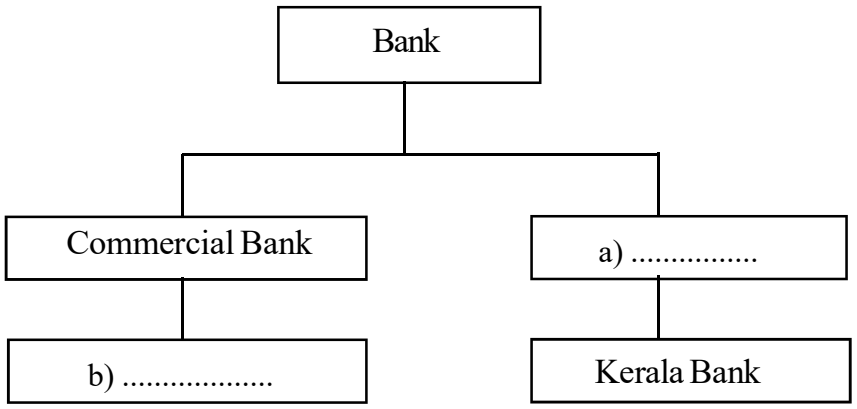
15 . Compare Fiscal policy And Monetary policy?

16. Complete the table given below.

(Nasik, Mumbai, Noida, Salboni)

Currency note Printing Press	Mint coins

17. Complete the flow chart given below



18. Commercial banks that were allowed to operate in India after the financial reforms of the 1990's are known as new generation banks. Write any two examples.

    a) Yes Bank b) ..... c) .....

19. The difference between the interest paid to depositors and the interest charged from borrowers is the income of banks.

    a) What is the income of the banks earned in this way called?
    b) What type of mortgages do banks accept and lend?

20. What are the other services provided by the Commercial Bank to the Public?

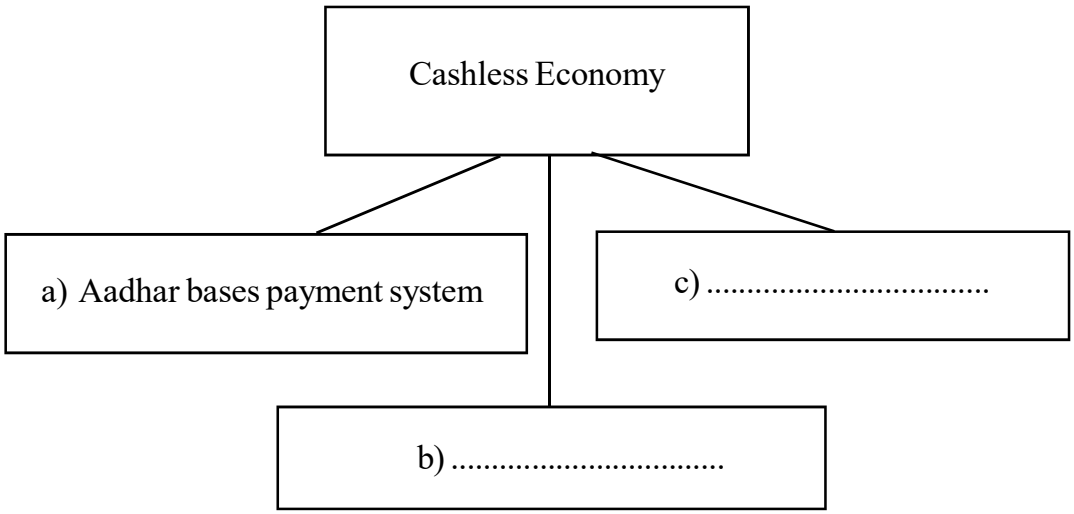
21. What is the facility that allow an account holder of a bank to carry out financial transactions from any of its branches ? How was this possible?

22. A man in Bangalore urgently needs to send money to his mother;but does not have time to go to the bank. Suggest any two transaction method?

23. Write the names of the regulatory institutions for non – banking financial institutions?

    a) Reserve Bank of India
    b) .....
    c) .....

24. Complete the Diagram



3 Mark

25. Look at the picture.



This picture is related to which function of money? List out and Explain other functions of money?

26. I am a 50 rupee note. At 8.30 am today , student took me to a shop and bought two notebooks worth 25 rupees each. Through many transactions I would have undergone by 5 pm, fully utilising my value.

Answer the following questions based on the explanation above.

a) What is the Velocity of circulation of money?

b) What changes occur in the economic system when velocity of circulation of money increases and decreases?

27. An increase in the general price level of goods and services is known as inflation. Answer the Questions given below about Inflation.

a) What causes Inflation?

b) What is used to calculate inflation in India?

c) Which institution prepares this?

28. Compare repo rate and reverse repo rate?

29. Term deposits are one of the deposits offered to the public by commercial banks. List out and explain other types of deposits?

30. A business man makes hundreds of transactions every day. Sometimes he has to withdraw more than the amount in his account. What type of deposit accounts is best for him? Why?

.31. Ramu plans to deposit some money in the bank for future needs. Suggest which investment account is suitable for him. Why?

32. A person takes a loan from a financial institution by pledging gold. But money can not withdraw by using check.

Answer the following questions based on the above features?

a) What financial institution is this?

b) Write another feature of these financial institutions?

c) Give another example of these financial institutions?

33. In order to bring the functioning of banks to different parts of the country and to more people,banks were nationalized. What are the main objectives of bank nationalization?

34. Co–operative banks play a crucial role in activating the rural economy by providing banking facilities to villagers and ordinary farmers.

a) On what principles does it operate?

b) Write down two objectives of co-operative bank?

35. Kudumbashree in Kerala encourages woman to take up employment by providing loans. Answer the following questions based on the above statement .

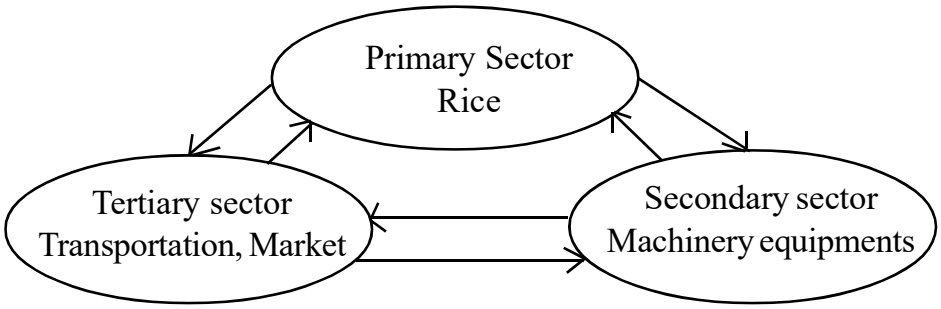
a) On What concept does Kudumbashree works?

b) Write two objectives of this?

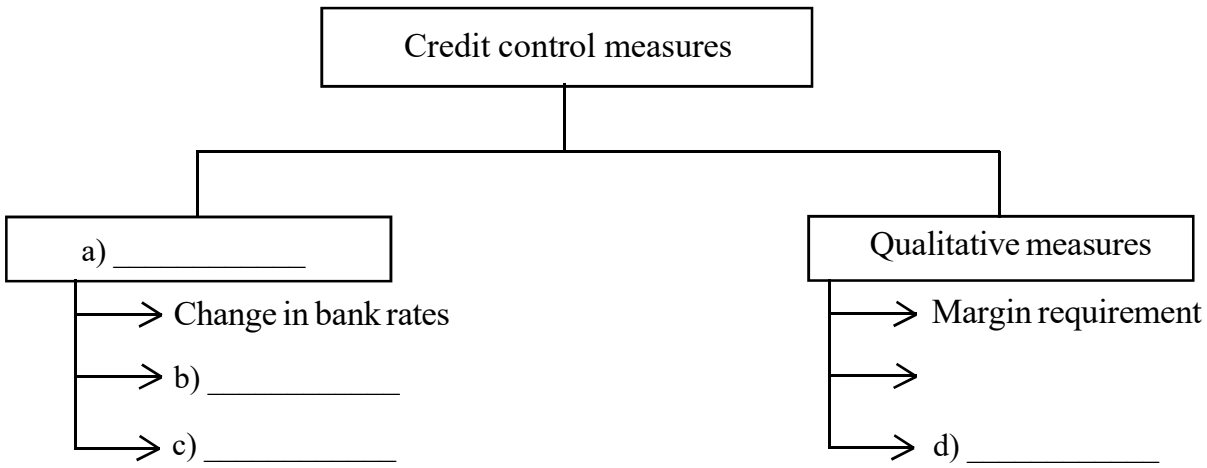
36. Suggest a scheme that would be useful for Raju to open a bank account without any minimum amount? Write down its main aims?

4 Mark

37. Explain how does money works in the economic system based on the diagram below.

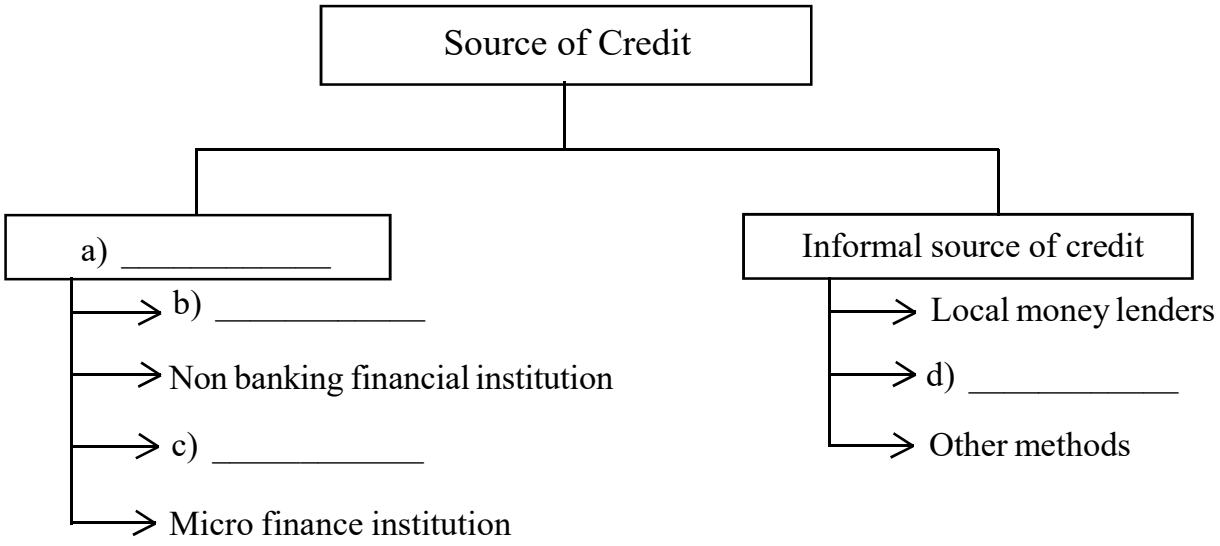


38. Complete the chart related to Credit control measures.



39. Teacher asked Manu to prepare a seminar on the topic of "use of banking technology". Help Manu write about the financial system that help speed up transactions through banking technology.

40. Complete the flow chart.



41. The source of credit in Kerala during various periods are given in the table.

Source/Period	2019 January-April	2020 January-April	2021 January-April
Banks	68.55	48.77	44.51
Non-bank financial institutions, Microfinance institutions	6.29	21.8	13.33
Self-help groups	8.8	17.43	41.53
Local moneylenders	8.04	3.65	4.43
Friends, relatives	7.85	6.79	2.99

Look at the table and answer the questions below.

a) Find out the reason for the decrease in loan banks during 2019-21.

b) What is the reason for the increase in debt of self help group during 2019-21?

c) Write about the influence of local money lenders on the credit system in Kerala?

5 Mark

42. The Central bank regulates and coordinates the activities of banks and non -banking financial institutions in the economy. Explain the functions of the Reserve Bank of India.

Chapter 5

MONEY AND ECONOMY

Answers

1 Mark

1. c
2. a
3. b
4. a
5. a
6. a
7. b
8. b
9. a
10. a

2 Mark

11. a) Exchange of goods and services

12. b) Use that money to buy another goods.

13. RBI
    It provides emergency loans to banks in the times of crisis, maintains the reserves of banks, and helps to settle transactions between banks.

14. M3 = M1 + net fixed deposits in commercial banks.
    M4 = M3 + total deposits in post offices(excluding National Savings Certificates)

15. Fiscal policy : It is the policy regarding taxation and government spending.
    Monetary policy : It is the policy regarding the supply of money and the rate of interest.

16. Currency note printing press: Nasik, Salboni
    Mint coins : Mumbai , Noida

17. a) Co-operative banker
    b) Public sector bank / private sector banks / small finance bank / payments banks / specialized banks /regional rural banks.

18. b) Axis bank
    c) Mahindra bank

19. a) Spread
    b) Gold, land documents, salary certificates

20. Credit card, Debit card
    ATM services, Locker facility

21. Core banking
    Core banking is a system that enables an account holder of a bank to carry out financial transactions from any of its branches. It is convenient for the customers.

22. Google pay , Paytm , Phone Pay , BIM UPI , and Amazon Pay

23. Securities and Exchange Board of India(SEBI) , Insurance Regulatory and Development Authority of India(IRDA)

24. b) e - wallet
    c) National financial switch

3 Mark

25. Measure of value : The value of all goods can be expressed in monetary terms.
    In the barter system, it was not easy to compare the value of one good with the value of another. Money made it easy to compare the values of two goods. Just as the value of goods is measured in monetary terms, the value of money can also be expressed in terms of other goods. The value of money is its purchasing power.

Store of value

It had become possible to store the value of any good in the form of money. This was not possible in the barter system. Through this it is possible to convert the value of goods, including that of perishable items, into money or asset and it can be used in the future.

Means of Deferred Payments

In the modern times many business activities are carried out with ease because of the advantage of settling the financial transactions at a later date. Both Buyers and sellers generally agree that the cash settlement of the transactions of goods and services can be cleared later.

26. a) The number of times a unit of money is exchanged in a given period of time is known as the velocity of circulation of money.
    b) An increase in the velocity of circulation money indicates an acceleration in economic growth,while a decrease in the velocity of circulation money indicates a slowdown in economic growth.

27. a) An increase in the general price level of goods and services is known as inflation.
    b) Inflation in India is measured using the Consumer Price Index.
    c) Which is prepared by the National Statistical Office under the ministry of Statistics and Programme implementation.

28. Repo rate : The rate of interest charged by the Reserve Bank of India on the loans taken by commercial banks from the RBI.
    Reverse Repo Rate : The rate of interest given by the Reserve Bank of India on the deposits by the Commercial Banks.

29. Savings deposits, Currents deposits, Recurring deposits

30. Current deposits: A current account is an account intended for business transactions. There is no limit to the number of transactions that can be made from such accounts in a single day. Banks do not pay interest on the money in this account. Overdraft facility is provided for this account. An overdraft is a system that allows you to withdraw more than the amount in the current account within a predetermined limit.

31. Term Deposit or Fixed Deposit Account : Money that is not needed to be withdrawn immediately can be deposited in such accounts. Banks pay more interest on such deposits than on money in a savings bank account. If money is withdrawn from these deposits before the maturity period, the interest rate received by the depositors may be reduced. The interest can be withdrawn upon maturity along with the deposits, or at various periods determined by the depositor.

32. a) Non – Banking financial institutions .
    b) Unlike banks, non – banking financial institutions cannot accept savings and deposits from the public.
    c) Insurance companies (eg . LIC ,GIC) , Mutual fund companies (eg UTI) ,and non – banking financial institutions(eg. KSFE)

33. 14 Banks were nationalized in 1969 and 6 banks, in 1980.

Main objectives :

* To expand banking facilities in rural areas
* To provide credit to farmers at lower rates
* To ensure equitable distribution of credit
* To prevent the concentration of economic power in a few people.

34. a) They operate on the principles of co -operation, self – help and mutual assistance.
    b) Inculcate the habit of saving among the villagers
    To protect the common people from private moneylenders
    To provide low -cost loans to farmers and small businessmen.

35. a) Microfinance
    b) Poverty alleviation , empowerment of women and the marginalized , promotion of entrepreneurship and ensuring job creation, and improvement of quality of life.

36. Jan Dhan Account
    Provide financial services to the low -income group
    Promote financial literacy

4 Mark

37. Rice production is an economic activity in the primary sector. A farmer who produces rice may need fertilizers , seeds, pesticides,machinery, and labour. When money is spent on fertilizers and machinery, it becomes a source of income for the producers in the industrial sector. When the rice producers depend upon transportation facilities to get the rice to the market, it becomes a source of income for service providers.

38. a) Quantitative measures
    b) Change in reserve rates
    c) Open market operations
    d) Moral suasion

39. National Electronic Fund Transfer System (NEFT)
    This is a system introduced by the Reserve Bank of India to make bank transactions between account holders easier and faster. Funds are transferred using the Indian Financial System Code(IFSC).

    Real Time Gross Settlement(RTGS)
    RTGS is a system introduced by the RBI to transfer large amounts of money between account holders. The feature of this is that transactions can be completed in a very short time. Core Banking Core banking is a system that enables an account holder of a bank to carry out financial transactions from any of its branches. There is no need to go to the specific branch where the customer holds the account for the bank transactions. It is convenient for the customers.

    Universal payment Interface(UPI)
    UPI is a payment system developed by the National Payments Corporation of India. It enables realtime money transfers between bank accounts. Users can connect their various bank accounts through a mobile applications and make simple and secure transactions. Some of the popular UPI apps are Google Pay, Paytm , Phone pay , BIM UPI and Amazon Pay.

40. a) Formal Source of Credit.
    b) Banks
    c) Self help groups
    d) Friends and Relatives

41. a) Loan rates for self help groups are increasing compared to banks.
    b) Local money transactions are not organized, informal, and non -institutional source of credit system of kerala .

5 Mark

42. Printing and issuing currency : As per the Reserve Bank of India Act 1934, only the Reserve Bank of India has the power to print and issue all currencies except coins and

one rupee notes. The coins and one rupee notes are printed and issued by the Ministry of Finance, Government of India. The Reserve Bank of India is responsible for designing, incorporating the security features, printing, and distributing the currency. Currency notes are printed at the Government of India's printing presses at Nasik (Maharashra) and Dewas (Madhya pradesh). It is also printed at two presses at Mysore (Karnataka) and Salboni (West Bengal) by Bharatiya Reserve Bank Note Mudran Limited, owned by RBI.Coins are minted at the Government of India's mints at Mumbai, Hyderabad,Kolkatta and Noida.

Banker's Bank : It provides emergency loans to banks in the times of crisis, maintains the reserves of banks, and helps to settle transactions between banks.

Controls the Supply of Money and credit : When the supply of money increases and the production of goods and services doesnot increase proportionately, there arises a situation where there are fewer goods and services and more money in the economy. This causes the prices of goods and services to rise.An increase in the general price level of goods and services is known as inflation.

Inflation in India is measured using the Consumer Price Index. Which is prepared by the National Statistical Office under the ministry of Statistics and Programme implementation.

Act as the government's bank : The Reserve Bank of India is responsible for maintaining government accounts, providing banking services, and implementing financial management. It also advises the government on matters such as fiscal and monetary policy.

Custodian of Foreign exchange reserves : The foreign exchange reserves of our economy are the sum total of foreign currencies and gold reserves. RBI is the custodian of all these.

Publication of Reports : RBI publishes various reports at different periods such as Banking Trends in India, Monetary policy Reports, Consumer Surveys,RBI Bulletin and statistical Supplements.

Chapter 6

THE CHANGING EARTH

1. Is the surface of the Earth where we live as a flat as a plain? Why the Earth's surface has so many ups and downs?

2. What is Earth movements? How they are classified? Explain

3. Which of the following does not cause Exogenic movements?
   i) Running water
   ii) Rotation of Earth
   iii) Variation in geothermal gradient
   iv) Slope of the area
   a) ii, iii b) i, iv
   c) iii, iv d) i, iii

4. Which of the following caused endogenic movements?
   a) Variation in geothermal gradient
   b) Gravity
   c) Rotation of Earth
   d) Radio activity within earth
   i) a, b, c ii) a, c, d
   iii) b, c, d iv) a, b, d

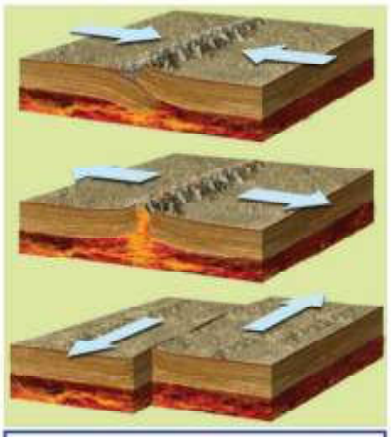
5. Observe the picture given below. Identify the geographical phenomenon to which each picture relates?



A

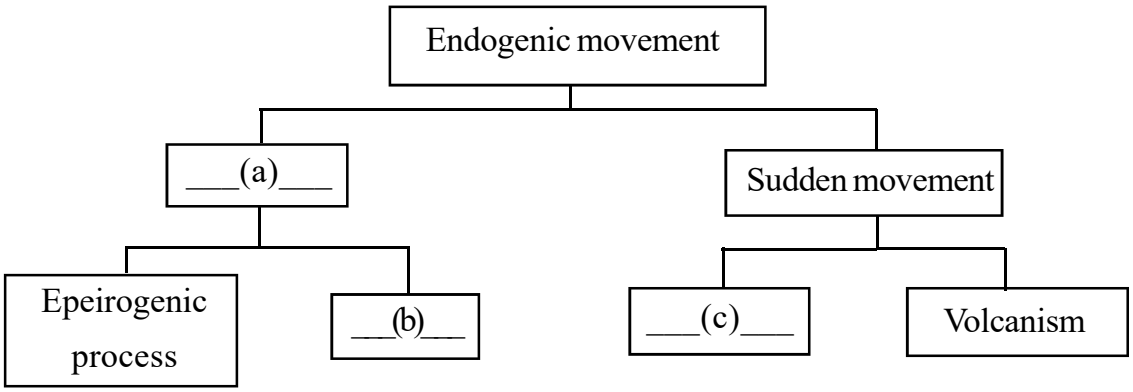


B



C

6. Fill in the appropriate places in the flow chart with the movements given below.



7. List out the difference between Epeirogenic movements and Orogenic movements?

8. Find out the sudden movements from the following?
   a) Earth quakes
   b) Volcanism
   c) Epeirogenic process
   d) Orogenic process
      i) a, c ii) a, b
      iii) b, c iv) a, d

9. What are the materials that erupt by volcanoes?

10. Find out the answers for the following statements?
    a) The molten material present in the upper mantle.
    b) Moten material reaches earth surface

11. Write a short note on Lithosphere?

12. What impact volcanic eruptions have on human life? Explain?

13. We can't prevent the eruptions of volcanoes. What scientific methods and precautionary measures we take to reduce the intensity of volcanic eruptions?

14. Still people are attracted in volcanic regions even though volcanoes are dangerous. Why?

15. There is an extensive area in india where soil has been formed from the disintegration of lava rocks. Which is that region?

16. Which is the only active volcano in India?

17. Why do lithospheric plates slide over the asthonosphere?

18. Volcanic activities are useful to human kind. Substantiate the statement?

19. Earth quakes are sudden movements. What are the causes of Earthquakes?

20. In tectonic earthquakes, the point inside the Earth from energy released is called focus or hypocentre. Name the point on the Earth's surface directly above the focus?

21. The Himalayan regions are highly vulnerable or susceptible to earthquakes. Why?

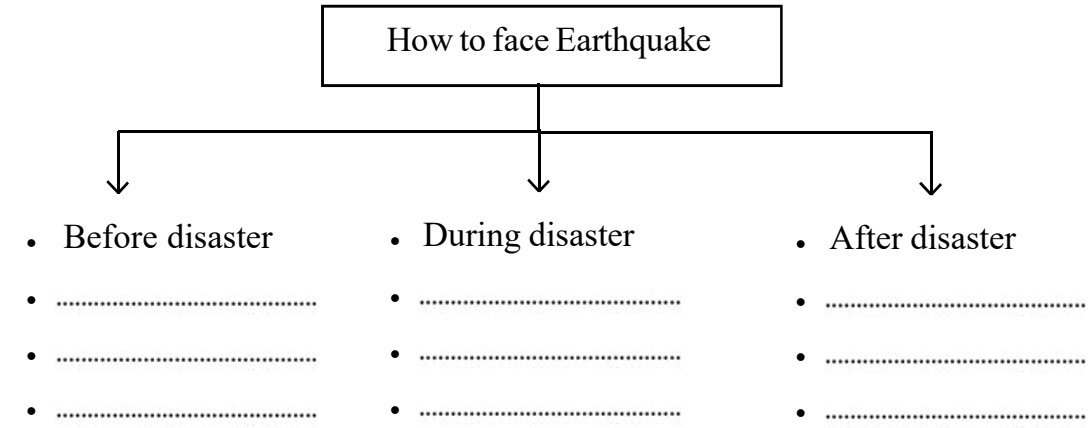
22. Which of the following is the unit to measure earthquake's magnitude?

    a) Pascal b) Celcius

    c) Richter scale d) Decibel

23. What impact earthquakes have on human life?

24. Prepare a chart titled 'How to face an Earthquakes' (before disaster, during disaster, after disaster).



25. The giant waves on ocean are called tsunami. What are the causes for tsunami?

26. What can we do to safeguard ourselves from tsunami? Explain

27. What are the Exogenic movements which cause ups and downs on earth's surface?

28. Weathering is one of the exogenic movements that cause ups and downs on the earth's surface. What are the forces that cause weathering?

29. Erosion, Transportation, deposition are the exogenic movements. What are the forces that cause these movements?

30. Mass wasting is one of the exogenic movements that cause ups and downs on the earth's surface. Which of the following cause mass wasting?

a) Running water
b) Gravitational force
c) Action of ground water
d) Temperature change

31. Because of certain exogenic forces, rocks are mechanically disintegrated chemically decomposed. The process that bring about these chemical and mechanical change in rocks are called

a) Mass wasting
b) Erosion
c) Weathering
d) Deposition

32. What is weathering? What are the different types of weathering?

33. Do human activities cause weathering? List human activities that contribute to the weathering of rocks?

34. Explain how can weathering be useful to us?

35. Write a short note on mass wasting?

36. What is mass wasting? What are the different types of mass wasting?

37. The sudden movements of rok debris and soil mixed with water down a steep slope under the direct influence of gravity is called

a) debris flow
b) Land falls
c) Land slides
d) Land subsidence

38. What are the possible factors that can lead to urulpottal in an area?

39. How to face urulpottal at the time of disaster and after disaster?

Chapter 6

THE CHANGING EARTH
Answer Key

1.
- The Earth's surface is constantly subjected to many changes.
- Numerous movements and resulting process have been occuring on the surface as well as within earth.
- Mountains, plateaus, valleys, rivers and so on were all created due to continously occuring Earth movements and process.

2.
- The movements that cause ups and down on the surface of Earth are called the Earth movements.

Earth movements can be categorised in to two types.

1. Endogenic movement 2. Exogenic movement

Endogenic movement

- The process that are cause by the forces within the earth at varied intensities.
- Radiouctivity within the earth, primordial heat the rotation of earth, variations in geothermal gardient and convection currents are causative forces for Endogenic movements.

Exogenic movements

- The process that are caused by external forces on the surface earth at varied intensities.
- Slope of the area, gavity, pressure exerted on an object by another object, molecular stress due to temperature change, running water, glaciers, wind, ground water and different chemical actions are causative forces for Exogenic movements.

3. a) ii, iii

4. ii) a, c, d

5. A. Earthquakes
   B. Volcanism
   C. Plate tectonics

6. a) Diastrophic movements
   b) Orogenic process
   c) Earthquakes

7. **Epeirogenic Process**
   - Process that cause the upliftment of a large portion of Earth's crust
   - Cause only minimal deformation

   Organic Process
   - Process that cause formation of mountains
   - Earth surface is severelly deformed

8. ii) a, b

9. Moten rock material, gases, ash pyroclastic material, dust and water vapour

10. a) Magma
    b) Lava

11. - Outer most layer of Earth
    - Crust along with upper portion of mantle
    - Solid state with an average thickness of 100 kilometres
    - Zone lying just beloow the solid lithosphere

12. * Loss of lives and destrull properties
    * Disrupt economic activities and severely affect human life
    * By destroying houses and land, it leaves many people homeless and jobless

13. - Avoid the construction of houses in areas close to volcanoes, restricting tourism in such area.
    - Continuously monitoring volcanic regions using scientific instruments and issuing early warnings.
    - Constructing building capable of withstanding lava flow and ashfall after eruption.

14. • The ash from volcanoes and the soil formed by disintegration of rocks created from solidified lava are very fertile
    • Such places are suitable for agriculture and hence attract people to settle there.

15. Deccan plateau

16. Barren Island

17. Asthenosphere is partially in molten state or semi-liquid state

18. • Lava soil is suitable for agriculture
    • The geothermal energy released from volcanic region is converted into electric ity in certain areas and utilised for loca purpose.
    • Lava is rich in minerals and many valuable minerals reach the surface of Earth through the flow of lava.
    • Volcanic regions often attract a large numbers of tourist.

19. • Lithospheric plate movements
    • Volcanic eruptions
    • The collapse of mines roots in intensive mining areas
    • Pressure exerted by large reservoirs

20. Epicentre

21. • Himalayan region are located on highly active, convergent tectonic boundary
    • Indian plate is continuously colliding with and thrusting beneath the Eurasian plate
    • Studies show, that the height of the Himalayas is increasing

22. c) Richter scale

23. • Killed thousands of people
    • Destroyed hundreds of building
    • Transportation and Communication system were shattered and water distribution failed

- Food supply was disrupted and the shortage of medicine was severe.
- Rescue operations made difficult by landslide and fires

24. **Before the disaster**
    - Secure your space
    - Ensure the structure of the house
    - Prepare Emergency kit (tourch, water, medicine)

    During the disaster
    - Stay safe
    - Don't use life
    - Take shelter under the chair or table

    After Disaster
    - Help injured
    - Follow the instruments of the rescue force
    - Check for hazards

25. - Earthquakes occur on the ocean floor
    - Volcanic eruptions on the ocean floor
    - Fall of meteors

26. - If the sea leve suddenly goes down by several metres, consider it a tsunami warning. Run away from the shore immediately and move to higher ground.
    - If you are caught in tsunami, try to save yourself by holding on to something that can float.
    - Follow the instructions of rescue force
    - Prepare emergency kit

27. - Weathering
    - Erosion, Transportation, deposition
    - Mass wasting

28. - Gravitational force
    - Temperature changes
    - Molecular stress
    - Bio-chemical actions

29. The Kinetic energy of running water, glaciers wind and waves and the action of ground water.

30. b) Gravitational force

31. c) Weathering

32. Because of certain exogenous forces, rocks and mechanically disintegrated or chemically decomposed. The processes that bring about these chemical and mechanical changes in rocks are called weathering.

    a) Physical weathering
    Due to temperature change or water gets trapped inside the cracks of rocks and freezes into ice, gradually breaks the rocks a part.

    b) Chemical Weathering
    Oxygen, Carbondioxide, water and other substances engage in chemical relation with minerals in rocks, leading to decomposition of rocks.

    c) Biological weathering
    The protrustion of plant-roots, burrowing animals and decay of plant and animal remains cause disintegration of rocks.

33. Yes
    - Mining, Quarrying
    - Deforestation, Construction
    - Changes in agricultural methods

34. - Formation of soil
    - Seperate minerals from rocks
    - It cause mineral enrichment
    - It makes mining easier

35. - In addition to weathering and erosion, another type of earth movement that occurs on the surface of earth due to exogenic forces is mass wasting
    - It refers to the movement or tall of rock tragments or earth material down a slope under the direct influence of gravitational pull.
    - It ranges from very slow movements to rapid movements
      Eg: Urul pottal

36. The phenomenon of fall or movement of rocks or topsoil from higher to lower areas under the direct influence of gravity is called mass wasting.

Different types of mass wasting

- Land falls
- Rock falls
- Land slide
- Debris flow
- Land subsidence

37. a) Debris flow

38.
- Slope of the land
- Depth of soil
- Structure
- Land use
- Distribution of streams

39.

At the time of disaster	After disaster
• Stay calm and do not panic • During urulpottal, if you are inside your house, take shelter under a strong table or cot • Take precautions to avoid head injuries • Do not seek shelter under trees	• Allow Ambulance and other rescue vehicles to pass without obstruction • Report immediately to the authorities if you notice any fallen electric lines in urulpottal affected regions • Do not visit the urulpottal region • Do not obstruct rescue workers

Chapter 7

INDIAN ECONOMY GROWTH AND TRANSFORMATION

1. What are the goals for which individuals, institutions and the govenment engage in economic activities?

2. Read the statements below and find the correct answer.

   Statement 1 : The Goods and Services are produced as a result of activities in the agricultural, industrial and service sector of an economy.

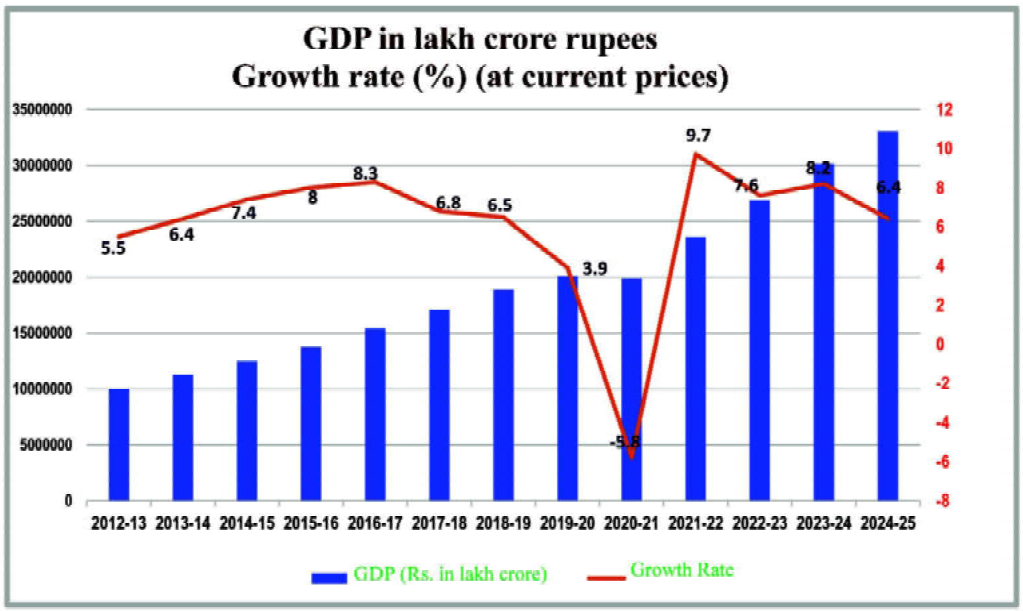
   Statement 2 : When the production of goods and services increases, the national income will increase and the economy will grow.

   a) Statement 1 and 2 are correct
   b) Statement 1 and 2 are false
   c) Statement 1 is true and 2 is false
   d) Statement 1 is false and 2 is true

3. What is the economic growth? How does economic growth benefit the economy?

4. Write the equation to find the economic growth rate.

5. The graph below shows the economic growth rate of India from 2012-13 to 2023-24. Analyse the graph and draw conclusions.



6. What is Percapita Income and how is it differ from personal income?

7. What are the limitations of percapita income as an indicator of a country's economic growth?

8. Find out the percapita income of the lower middle class that includes India, as per the development report prepared by the World Bank.

   a) Less than 1145
   b) Above 14006
   c) 4516 to 14005
   d) 1146 to 4515

9. Which of the following is not included in the factors used to prepare the Human Development Index?

   a) Life Expectancy
   b) Infant mortality rate
   c) Literacy and gross school enrolment rate
   d) Percapita income

10. The countries of the world are classified into four categories based on the value of the Human Development Index. Which of the following has the highest human development?

    a) Above 0.800
    b) 0.550 - 0.699
    c) 0.700 - 0.799
    d) Below 0.550

11. Write the equation to find the physical quality index. Explain what each of the indices given in the equation means.

12. 'Quantitative change refers to economic growth and qualitative change refers to economic development' - Compare Economic development and economic growth based on the statement.

13. The World Development Report published by World Bank devides economics into four categories based on percapita income - What are they?

14. The main shortcomings of an economic growth index is given below. Identify the economic growth index and list the other shortcomings.

    * It is only an average income

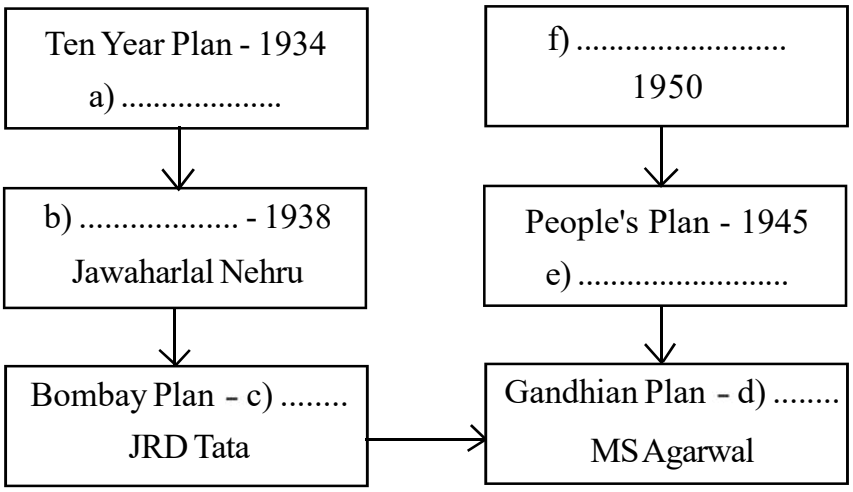
15. To improve the quality of life, economic growth must be accompanied by better living conditions. Find out what such conditions are?

16. The Human Development Index is calculated based on three factors. One among them is given below. Find the factors.

    * Life expectancy

17. On the basis of which indicators is the Physical Quality of Life Index prepared?

18. Complete the flow chart showing the growth planning in India.



19. Examine the table given below and answer the questions given based on the HDI Value.

Year	HDI Value
1990	0.446
1995	0.469
2000	0.497
2005	0.547
2010	0.590
2015	0.633
2020	0.652
2023	0.685

a) During which years was India at a low stage of Human Development?

b) In which years did India achieve medium human development?

c) Findout the areas in which India need to make progress to achieve high human development?

20. Which of the following statements related to the Sustainable Development Index is correct?

Statement 1 : Sustainable Development Index is a comprehensive development index.

Statement 2 : This is a development index prepared by NITI Ayog in collaboration with the United Nations.

Statement 3 : This is related to Physical Quality Index

a) Statement 1, 2, 3 are correct

b) Statement 1 and 2 are correct

c) Statement 1 and 3 are correct

d) Statement 2 and 3 are correct

21. Which of the following is the development index developed by Morris David Morris?

a) Human Development Index

b) Physical Quality of life Index

c) Sustainable Development Index

d) Percapita Income

22. What is the main drawback of the Physical Quality Index?

23. There are 17 goals in the sustainable development goals. Write any three environmental goals?

24. Responsible Consumption and Production are Sustainable economic goals adopted by the UN. What are the other goals included in this?

25. Some indicators related to sustainable development are given below. What does this score indicate?

Score	Category
0 - 49	
50 - 64	
65 - 99	
100	

26. Why is Kerala leading in the SDG index compared to other states in India?

27. Who is considered as the father of Indian economic planning?

28. The objectives of Indian economic planning are given below. List what they aim to achieve.

Objectives	
Growth	
Modernization	
Self-reliance	
Equality	

29. Which of the following statements regarding the Planning Commission is incorrect?
    a) Chairman - Prime Minister
    b) The Planning Commission came into existence in 1951
    c) Its members are appointed by Central Government

30. Who is considered as the architect of the Five Year Plan in India?

31. In which of the following year did NITI Ayog come into existance?
    a) 2023 Juy 1
    b) 2015 January 1
    c) 2019 May 1
    d) 1950 January 1

32. What are the objectives of NITI Ayog?

33. The Chairman of NITI Ayog will be the Prime Minister and the Vice-Chairman will be a person nominated by the Prime Minister. Who are the other members of it?

34. One of the main objectives of NITI Ayog is to promote sustainable development and inclusive growth. What other objectives does it have?

35. What methods did the Government adopt to control the private sector before the 1990's?

36. What were the circumstances that prompted India to switch from LPQ to LPG?

37. What were the main objectives of the new economic policy?

38. Explain what LPQ and LPG are? LPQ, LPG

39. What are the three basic elements of the new economic policy?

40. In which sectors did India implement reforms as part of Liberalisation?

41. How did disinvestment help privatisation?

42. The liberalisation and privatisation are two economic policies. The concequence of these policies is called Globalization. Why?

43. Globalization is not a policy; It is a phenomenon - Who said this?

44. Write any two results of Globalization?

45. The Industrial Revolution that began with the invention of the steam engine is now on the path of the fifth industrial revolution. List the characteristics of the Industrial revolution that have occured to date.

46. How do the economic activities in a knowledge economy lead to economic growth?

47. What changes have been made in the Indian Agriculture Sector in the new world order based on the knowledge economy?

48. Why is it said that the knowledge economy is making the Indian Industrial Sector SMART?

49. How do the technological advances in the service sector help the Industrial Revolution?

50. List out the instructions of knowledge based economy? What are the objectives of these instructions?

51. What are the activities of K-DISC, which works with the aim of transforming Kerala into a knowledge economy?

52. How does the Kerala Startup Mission promote innovative ventures?

•

Chapter 7

INDIAN ECONOMY GROWTH AND TRANSFORMATION

Answers

1.
   - Increase in infrastructure
   - Better job opportunities
   - Increased income and purchasing power
   - Higher Quality of Life

2. a) Statement 1 and 2 are correct

3.
   - Economic growth is the increase in a country's Gross Domestic product compared to the previous year.
   - When economic growth occurs, a country's capasity to meet the needs of its people increases through changes in production, employment and income.

4. Economic Growth rate = $\frac{\text{Current Year's GDP - Previous Year's GDP}}{\text{Previous Year's GDP}}$ X 100

5.
   - The production of goods and services does not necessarily increase steadily every year.
   - GDP is increasing although the growth rate has showed 2016-17 to 2019-20.
   - From 2019-20 to 2020-21 GDP and Growth rate declined and economic growth turned negative.
   - From 2020-21 to 2021-22 GDP and Growth rates increased and economic growth shifted from negative to positive.

6.
   - Percapita income is the average income of the people.
   - Percapita Income = $\frac{\text{National Income}}{\text{Population}}$
   - Persona income is ones actual income

7.
- It is only an average income
- Information about the distribution and inequality of wealth is not reflected in the income index.
- Since Percapita income is calculated on the basis of national income, the limitations of calculating national income also affect Percapita income.
- Factors that enhance the quality of life are not included in the scope of the Percapita income index.

8. d) 1146 to 4515

9. b) Infant Mortality rate

10. c) 0.700 - 0.799

11. $PQLI = \frac{LEI + IMI + BLI}{3}$

LEI : Life Expectancy Index

- Life expectancy is calculated based on the average number of years a person lives.

IMI : Infant Mortality Index

- Infant Mortality rate refers to the number of deaths of infants under the age of one per 1000 live births.

BLI : Basic Literacy Index

- Basic Literacy is the percentage of the population that can read and write.

12. **Economic Growth**

- Quantitative Change
- Increase in production of goods and services
- Increase in National Income
- The Country's Capasity to meet the needs of the people is increasing.
- Growth in GDP

Economic Development

- Qualitative change
- There is a change in the quality of life and well being of people.
- Improving Financial Well-being

- Improving Welfare activities
- Economic development is measured in terms of development indices.

13. 1. Low income
    2. Lower middle income
    3. Upper middle income
    4. High income

14.
- It is only an average income
- Personal income is one's actual income
- Percapita income may be higher or lower than individual income
- There is difference between Per capita income and Personal income.
- Information on economic distribution and inequality is provided by per capita income. Not reflected in dimension.
- Since Per capita income is calculated on the basis of national income, the limitations of calculating national income also affect Per capita income.
- Factors that enhance the quality of life are not included in the scope of the Per capita income index.

15.
- Educational facilities
- Nutrient availability
- Transportation facilities
- Medical facilities
- Housing facilities

16.
- Life expectancy
- Literacy and gross school enrollment rate
- Per capita income

17.
- Basic Literacy Index
- Infant Mortality Index
- Life expectancy

18. a) M. Visvesvaraya
    b) National Panning Committee
    c) 1944
    d) 1944
    e) M.N. Roy
    f) Planning Commission

19. a) 1990, 1995, 2000
    b) 2010, 2015, 2020, 2023
    c) Life expectancy, Education, Improved Quality of Life.

20. b) Statement 1 and 2 are correct

21. ii) Physical Quality of Life Index

22. Per Capita is not considered

23. • Climate change
    • Protection of underwater life
    • Conservation of life on land

24. • Affordable and Clean energy
    • Decent work and economic growth
    • Industrym, innovation and infrastructure

25. 0 - 49 - Aspirant
    50 - 64 - Performer
    65 - 99 - Front Runner
    100 - Achiever

26. • Great Strides in the areas of education, health and social justice.
    • High Literacy rates
    • Low Infant Mortality rates
    • Improved Life Expectancy
    • Land Reform
    • Public Distribution System
    • Social Security Schemes

27. M. Visvesvaraya

28. Growth :- An Increase in production

    Modernisation :- Use of new technologies and ideas and modernising social perspectives.

    Self-reliance :- Minimise foreign dependence on economic activity.

    Equality :- Ensuring equality by providing basic needs to all and enabling fair distribution of wealth.

29. b) The Planning Commission came into existence in 1951.

30. P.C. Mahalanobis

31. b) 2015 January 1

32. • Economic Growth
    • Social Welfare
    • Infrastructure development

33. All Chief Ministers and Lieutenant Governors and Union Teritories as members.
    Four Union Ministers and Non Official members as nominated by the Prime Minister.

34. • To create a vision for India's development priorities and plans.
    • Promote sustainable development and inclusive growth.
    • To promote economic growth through innovation, entrepreneurship and job creation.

35. Through Licenses, Permits and quotas (LPQ)

36. • High fiscal deficit
    • Decrease in foreign exchange reserves
    • Public sector inefficiency
    • Terms of world and IMF
    • The economic crisis caused by the Gulf war

37. • Liberalise Indian economy
    • Reduce Restrictions in the private sector
    • Remove Government regulations

38. LPQ - License, Permit, Quota

LPG - Liberalisation, Privatisation, Globalisation

39. Liberalisation, Privatisationand Globalisation

40. Industrial Sector, Financial Sector, Tax Reforms, Foreign Exchange Reforms, Foreign Trade Sector

41. Dis investment is the sale of Government's investment or capital in a public sector enterprises to the private sector. This is mainly done in four ways.

Wholesale, Share sale, Capital sale, Strategic sale

42. Globalisation is the result of policies that have made economics more interconnected and integrated. With this, global trade becomes regulated by the world trade organisation.

43. Amartya Sen

44. Foreign Capital Investment increases

Encouraging research and innovation

45.

Industrial Revolution	Features
1. Invention of the steam engine:	• Emergence of the factory system • Changes in production and transportation
2. Electricity	• Use of electricity in the place of steam. • Expansion into new areas of production and distribution
3. Electronics & Automation	• Use of robotics and automation in factories
4. The era of advanced technologies	• The beginning of Internet of Things (IoT) sensors and Big Data Analytics, Cloud Computing, Nanotechnology
5. Human-centric approach to digital technologies including AI	• Sustainable productivity growth and expansion of smart factories through technologies like AI, IoT and Robotics.

46.
- Skill development. Utilisation of Human Resources
- Creation of employment opportunities
- Research and creation of innovations.

47.
- Making it more profitable and efficient (SMART)
- AI Technology is being used in Agriculture
- Low cast and high yield
- Sales through online platforms

48.
- Factories are connected using computerised Automation and internet.
- Increased industrial production
- Quality products
- Less Wastage
- Diverse job opportunity
- Project like Make in India can be implemented

49.
Edu Tech - Robotics, AI, IOT
Health Tech - Tele medicine
Green Tech - Renewable Energy
Animation - Advertisement

50.
- Kerala University of Digital Science, Innovation and Technology. (Digital University of Kerala)
- Kerala Development and Innovation Strategy Counsil (K-DISC)
- Kerala Start up Mission (KSUM)

51.
- The committee was formed in 2018 with the aim of transforming Kerala into a knowledge economy and a hub of innovation.
- Promote research in new technologies, products and processes.
- Organise Skil Development Programmes.
- Foster innovation among the youth

52. To promote the start up scene in Kerala, transform innovative ideas into enterprises. It supports start ups through innovation hubs, incubators, funding support, mentorship and international exchange programmes.

Chapter 8

TOWARDS SUSTAINABILITY

1. Categorize and list the resources you are familiar with into natural and man made resources?

   (Minerals, National high way, Solar energy,
   School, Worship place, Coal)

2. Resources can be classified into different categories based on their origin .which are they?

3. Find the related resources that can be renewable in the given below
   a) These are sustainable resources as long as they are not overused
   b) These are resources that are formed over millions of years and diminish in quantity with use
   c) Examples are solar energy and wind
   d) Examples are coal and petrol

   a) a, b Related b) a, c Related
   c) a, d Related d) b, d Related

4. Explain the classification of resources on the basis of renewability?

5. On the basis of ownership, resources can be divided into several categories. Some examples are given below and list them to find out which resource they are related to,

   (Gardens, Mineral Wealth, Home, Marine Resources Beyond 200 Nautical Miles, Water Bodies, Public Playground, Own Vehicle, Forests)

6. Hydrogen is an example of stocks. explain what is stocks with the help of hydrogen?

7. Find out those which are related to titanium in the statements given below
   a) Haematite, magnetite, and limonite are ores of titanium
   b) Ilmenite and rutile are ores of titanium

c) Stronger than steel but only half the weight of steel,

d) It is used to make air planes, missiles, spacecraft, etc.

a) a, b, c related b) b, c, d related

c) b, c, related d) a, c, d related

8. What is the ore of Aluminium?

a) Magnetite b) Sphalerite

c) Bauxite d) Chromite

9. Below is information related to a major metal. Identify which metal it is, and explain its significance

- Caragas mine located in Brazil
- An important component of steel making
- It is mined in about 50 countries

10. Below are descriptions related to a major metal and identify which metal it is,

- It is an indispensable metal in the electrical industry
- Chile ranks first in production
- Early humans used it for decorative purposes, making tools and making weapons
- Escondida , Collahuasi are the important mines

11. What is fuel? Write an example for renewable fuel?

12. Classify the energy resources given below into conventional and non-conventional resources on the basis of renewable potential

(• Solar energy • Coal • Hydel power • Wind • Nuclear energy • Bio energy)

13. Compare the characteristics of both conventional and non-conventional energy resources.

14. The characteristics of a major fossil fuel are given. Identify the fuel and explain its uses.Formed from plant debris

* High organic nature

* available in black or brown colours

15. What are the products produced from petroleum?

16. What is meant by sustainable development?

17. State the importance of hydrogen as a traditional energy source.

18. Explain the major non-conventional energy resources, and how they can be utilized for our energy needs?

19. Explain the role of resource conservation in achieving sustainable development

20. In order to solve the energy crisis, it is imperative to expand the use of renewable energy sources. Substantiate

Chapter 8

TOWARDS SUSTAINABILITY

Answers

1.

Natural resources	Man made resources
* Minerals	* National Highway
* Solar energy	* School
* Coal	* Worship places

2. 1. Biotic resources

Anything that is part of the biosphere and can be used by humans is called abiotic resource.

Examples; humans, plants and animals .

2. Abiotic resources

Resources that are made of non living matter are called abiotic resources

Examples; minerals, wind

3. b) A, C related

4. Classified into two

1. Renewable resources
   - These are resources that are continuously produced or replenished in nature through physical or chemical processes.
   - Many of them do not diminish after use and can be reused. These resources are considered sustainable as long as they are not over used examples, sunlight, wind
2. Non renewable resources
   - Non renewable resources are those that have been formed over millions of years and whose reserves decrease with use examples; fossil fuels and Minerals

5.

Personal resources	Community resources	National resources	International resources
• house • own vehicles	• Park • public playground	• wildlife • forest • mineral deposit	• Marine resources beyond 200 nautical mile

6. Hydrogen can be utilised as a rich source of energy .But we have a not yet developed the advanced technology to use it as a resource today .Materials that can meet human needs but for which humans do not have the appropriate technology to utilise are called stocks.

7. d) a, c, d related
8. c) Bauxite

9. Iron
   - Iron is an indispensable metal in our daily life.
   - India is one of the major iron producing country in the world
   - About 98 percentage of the iron ore mined worldwide is used to produce Steel.
   - Steel place a crucial role in maintaining a strong industrial base.

10. Copper

11. Any substance that releases energy through chemical change can be called a fuel
    eg: solar energy

12.

Conventional resources	Non conventional resources
• Coal • Hydel power • bio energy	• sun light • wind • nuclear energy

13. **Conventional resources**
   - These resources get depleted and exhausted with use
   - They cause pollution
   - Very expensive to maintain

Non conventional resources

- Non conventional resources available in plenty
- Eco friendly
- Usually pollution free.

14. **Coal**
    - Coal is used as fuel in metallurgy
    - It is also used as chemical raw material in the manufacturing of waxes,ointments, medicine, pesticides and dyes
    - Coal is a major source of energy for thermal power generation

15. Paints, chemicals, synthetic fibres, fertilizers, lubricants, and thousands of other products are made from petroleum today.

16. Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

17.
    - Hydrogen a chemical element is used to produce energy
    - Which can serve as clean fuel
    - When hydrogen is burnt in a fuel cell water paper only released
    - The main peculiarity of this energy sources it does not a release pollutants like fossil fuels when they are burnt
    - This energy is useful in area ranging from transportation to various economic activities

18. **Solar energy**
    - Sun is the primary source of energy
    - Many techniques are currently in used to convert Solar rays into energy
      Example; photo voltaic and Solar thermal capture

    Wind
    - The kinetic energy of the wind is converted into electricity
    - The force of the windows used to spin a windmill. which in turn drives a generator connected to it producing electricity

Geothermal energy

- The heat released when Magma erupt from the Earth interior is efficiently stored and converted into electrical energy.
- It is possible to generate thermal energy from the hot water and vapour that emerge from the earth through geysers

Tidal energy

- The movement of ocean tide is spin turbines and generate electricity.

Energy from Hydrogen will play an important role in the energy supply in the coming days

19.
- Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.
- Utilize natural resources without damaging the ecosystem and impacting the global climate
- Develop solar, wind, hydropower and geothermal energy
- Reducing the use of fossil fuels
- Reduce overuse of natural resources

20. In order to solve the future energy crisis due to excessive resource consumption and the damage already done to nature, we need to focus on other sustainable energy sources instead of current energy sources.

That means we need to develop renewable energy sources and in this case we have to rely on non-conventional energy sources which are always available and sustainable.

We should be able to use solar energy, wind energy, geothermal energy, tidal energy, hydro electricity, biomass energy as our energy sources.

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